Robust Downlink Precoding for LEO Satellite Systems With Per-Antenna Power Constraints

Yanhao Liu, Yibiao Wang, Jue Wang, *Member, IEEE*, Li You, *Senior Member, IEEE*, Wenjin Wang, *Member, IEEE*, and Xiqi Gao, *Fellow, IEEE*

***Abstract*—Downlink precoding has been widely accepted as a crucial technique in low-earth-orbit (LEO) satellite communications due to its advantages in improving the space division multiplexing capability and enhancing spectrum efficiency. Because of the strong spatial directivity in the satellite channels, the downlink precoder can be efficiently designed based on the angle of departure (AoD)-based channel state information (CSI). However, acquiring perfect AoD-based CSI is challenging due to the angle estimation error and attitude jitter of the LEO satellites. In light of this, this paper investigates the precoder design for the LEO satellite downlinks with imperfect AoD-based CSI. For robust design, we treat the angle deviations of each user as random variables, which are supposed to exhibit a particular distribution, then integrate them within a certain range to achieve the robustness of the proposed downlink precoder against the angle mismatch. Instead of using the sum-power constraint on the transmit antennas, we adopt a more realistic per-antenna power constraint (PAPC). Accordingly, we formulate the problem to maximize the ergodic sum rate of the system. By leveraging the corresponding Lagrangian formulation and identifying the optimal precoder structure as the solution to a generalized eigenvalue problem, an iterative algorithm for the robust precoder design is derived. To reduce the computational complexity, a low-complexity version of the proposed algorithm is implemented based on deep learning techniques to meet the high demand for real-time applications in LEO satellite systems. Simulation results verify the effectiveness of the proposed precoding approach.**

***Index Terms*—LEO satellites, robust precoding, per-antenna power constraint (PAPC), imperfect CSI, angular mismatch, deep learning.**

I. Introduction

LARGE low-earth-orbit (LEO) constellations, including SpaceX's Ku-Ka-band LEO satellite system, OneWeb's Ku-Ka-band satellite system and Telesat's Ka-band satellite system, have attracted much attention from industry and academia due to their reliable services, vast coverage, lower production cost and lower latency compared with the medium-earth orbit (MEO) and geostationary-earth orbit (GEO) counterparts [1], [2]. To take full advantage of the spatial characteristics of the LEO satellite downlink channels, alleviate multiuser interference and improve the sum-rate performance, the downlink precoder for all served users for the satellite systems should be appropriately designed.

In the past decades, downlink precoding technology has received widespread attention [2], and the precoder design has been formulated as various problems, such as signal-to-interference-plus-noise-ratio (SINR) balancing [3], [4], sum-rate maximization [5], [6], and transmission power minimization [7]–[9], etc. In multi-beam satellite communications, the downlink precoder is typically designed based on the beam domain channel matrix [10], [11], which is obtained by performing feature decomposition on the traditional physical channel matrix [12]–[14]. This kind of precoding scheme can greatly reduce multiuser interference and computational complexity due to the non-overlapping beams assigned to different users and the reduced channel matrix dimension [14]. On the other hand, the downlink precoder design can also be performed in antenna domain [15]. The LEO satellites are usually equipped with uniform planar arrays (UPAs) because of the compact sizes, multiuser multiplexing, and three-dimensional (3D) coverage, making the antenna domain channels have strong spatial directivity, and the downlink precoder can be efficiently designed based on the angle of departure (AoD)-based channel state information (CSI). Compared with the beam domain design, antenna domain precoding can increase the available degrees of freedom for the LEO satellite equipped with a large number of antennas.

It is worth noting that the downlink precoding performance is susceptible to the accuracy of the AoD-based CSI available at the satellite. However, in LEO satellite communication systems, it is challenging to acquire perfect AoD-based CSI of each user due to practical factors, such as large propagation delay, high mobility, positioning angle error, and attitude jitter of the satellites [16]–[18]. In particular, the rapid variation of the satellite's position caused by the high-speed movement and the increased feedback time of the AoD information to the satellite because of the large transmission delay can make

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Yanhao Liu, Yibiao Wang, Li You, Wenjin Wang, and Xiqi Gao are with the National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China, and also with the Purple Mountain Laboratories, Nanjing 211111, China (e-mail: yanhaol@seu.edu.cn; wyb_kongxin@qq.com; lyou@seu.edu.cn; wangwj@seu.edu.cn; xqgao@seu.edu.cn).

Jue Wang is with the School of Information Science and Technology, Nantong University, Nantong 226019, China (e-mail: wangjue@ntu.edu.cn).

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the AoD-based CSI outdated. Besides, the positioning error of the current single-satellite positioning algorithm can reach about 10 km, making it almost impossible for the LEO satellite to obtain perfect CSI [18]. As the number of served users increases, the high-precision satellite positioning will bring greater power consumption to the satellite system, affecting the energy efficiency and service life of the LEO satellite [17], [18]. On the other hand, in the majority of existing approaches, the sum-power constraint on the transmit antennas has been assumed to simplify the analysis [3], [5], [6], which may cause non-linear distortion of the signals and extra power consumption because of the nonlinearity of the power amplifier [19], [20]. Thus, in practical implementations of the energy-efficient and power-limited satellite systems, it is better to constrain the transmit power of each transmit antenna individually. Based on the above facts, investigating the optimal downlink precoder design based on the imperfect AoD-based CSI with the angular mismatch of each user and under the practical per-antenna power constraints (PAPCs) is a meaningful and practical research direction for the LEO satellite communication systems.

Moreover, real-time performance and power consumption are crucial for the LEO satellites, which should be emphatically considered in the precoder design. However, existing methods for solving the optimal precoding problem usually rely on iterative algorithms [4], [21] or convex optimization [22], [23]. The iterative calculation process often contains high-dimensional matrix inversion or product, causing high calculation delay and power consumption. Thanks to the latest developments in deep learning (DL) techniques in wireless communications [24]–[28], a neural network can be trained offline and then deployed for real-time online optimization. This makes it possible to obtain the precoder with low computational complexity and high real-time performance [29], [30].

A. Prior Work

In conventional terrestrial communication networks, modeling of the channel uncertainty is essential for robust precoder design. This channel uncertainty is often modeled as an independently and identically distributed (i.i.d.) complex Gaussian random matrix for simplicity [31]–[33]. In LEO satellite communications, due to the sparseness of the angle-based CSI, the channel uncertainty can be characterized by AoD-based CSI with angle mismatch. Because of the difficulty of obtaining perfect AoD-based CSI, robust precoder design based on imperfect angle-based CSI has attracted attention recently [34]–[36]. [34] proposed a discretization method to discretize the channel uncertainty region based on angular information and achieved satisfying robustness for precoder design by combining the possible angle-based discrete channel samples. [35] investigated the 3-D massive multiple-input multiple-output (MIMO) for air-to-ground transmission and proposed a location-assisted two-layer precoding scheme, where the effect of AoD uncertainty on the performance was evaluated. [36] studied the robust secure beamforming (BF) using an angle-of-arrival (AoA)-based channel uncertainty model in a satellite-terrestrial network. However, to the authors' best knowledge, there are few reports on modeling of the angle-based channel uncertainty for LEO satellite communication systems, which can precisely describe the angular mismatch of the users' channels. Nevertheless, the angle estimation error and attitude jitter of the LEO satellites may still render uncertainty in its acquired AoD information. For this reason, it is necessary to develop a specialized model for the LEO satellite downlink channels with angular uncertainty.

Recently, the per-antenna power constraints problem has been systematically researched [19], [20], [37]–[39]. For example, the literature [19] first proposed a dual structure to transform the per-antenna transmitter optimization problem into an equivalent minimax optimization problem. The work in [20] investigated the per-antenna constant envelope precoding for massive multiuser MIMO systems. Further, [37] proposed a framework for the transceiver designs for multi-antenna multi-hop cooperative communications under PAPCs and investigated both the linear and nonlinear transceiver designs. Besides, [38] investigated the signal-to-leakage-plus-noise ratio (SLNR) maximized downlink precoding under PAPCs, for which a semi-closed form optimal solution was proposed. In view of the high computational complexity and lack of robust design, it is not feasible to directly apply the above-mentioned precoding methods for the LEO satellites [2], [10]. Thus, in the subsequent paper, we will investigate the robust downlink precoder optimization under practical PAPCs for the LEO satellite communication systems.

B. Contributions

This paper aims to address the problem of the robust downlink precoder design under PAPCs for the LEO satellite multiuser multiple-input single-output (MISO) downlinks with AoD uncertainty of each user. Our design objective is to maximize the ergodic sum rate of the system, and the contributions of this work are summarized as follows:

- We develop an angular information-based uncertainty channel model to describe the angular estimation error (or angular mismatch) in each user's AoD for the LEO satellite downlink channels.
- We further integrate the angle deviations of each user within a certain range to facilitate the proposed precoder to resist the influence of the angular mismatch, where the angular deviations are treated as random variables that are assumed to exhibit a particular distribution.
- Aimed at solving the ergodic sum-rate maximization problem under PAPCs, we transform it into an improved Quality-of-Service (QoS) problem by which the structure of optimal precoder is characterized based on the Karush-Kuhn-Tucker (KKT) conditions of the corresponding Lagrangian function.

  We show that the direction and power allocation of the optimal precoding vectors can be obtained according to the solution to a generalized eigenvalue problem. After that, an iteration algorithm is proposed to determine the Lagrangian multipliers.

- A low-complexity version of the proposed robust precoding approach is implemented to meet the real-time application requirements using the trained deep neural networks (DNN), effectively reducing the computational latency while guaranteeing satisfactory performance.

The remainder of this paper is outlined as follows. Section II presents the downlink channel model with angular deviations, the transmission signal model, and the problem formulation. In Section III, the design of the optimal robust precoder is investigated. In Section IV, a low-complexity version of the proposed solution is developed based on DNN. Numerical results are presented in Section V. The conclusion is drawn in Section VI.

Notations: Upper and lower case boldface papers denote matrices and column vectors, respectively. $\mathbf{I}$ represents an identity matrix and $\mathbf{0}$ represents an all-zero matrix, with the dimension being determined from the context. $\mathbf{1}_{M\times N}$ represents an $M\times N$ all-one matrix. $\mathbf{e}_n$ denotes the n-th column vector of the identity matrix. $\mathbb{C}^{M\times N}$ denotes the $M\times N$ dimensional complex matrix space. $(\cdot)^H$ and $(\cdot)^T$ represent Hermitian transpose and transpose, respectively. $\otimes$ denotes the Kronecker product. $\mathbb{E}\{\cdot\}$ denotes the expectation operation. $\mathrm{Re}(\cdot)$ and $\mathrm{Im}(\cdot)$ denote the operations of taking the real and imaginary of a complex number, respectively. $\lfloor z\rfloor$ represents the largest integer not greater than z. $(\cdot)_i$ denotes the i-th element of a vector and $[\cdot]_{i,j}$ denotes the (i,j)-th element of a matrix. $\sim$ denotes 'be distributed as'. $\mathrm{Tr}(\cdot)$ represents the matrix trace. The expression $U[a,b]$ represents the uniform distribution in the interval $[a,b]$, and $\mathcal{CN}(\mu,\sigma^2)$ represents the complex Gaussian distribution with mean μ and variance σ^2. The operation $\mathrm{proj}(\cdot)$ represents the projection of a matrix on the space of diagonal positive semidefinite matrices. In addition, when the operation $\mathrm{diag}(\cdot)$ is operated on a vector, it will produce a diagonal matrix with the elements of the vector on the diagonal, and when it is operated on a matrix, it will generate a vector composed of the diagonal elements of the matrix.

II. System Model

Consider a downlink transmission scenario where an LEO satellite serves K single-antenna users simultaneously. The LEO satellite is equipped with a UPA, which is composed of $N_T = M_xM_y$ antennas where M_x and M_y are the numbers of antennas on the x- and y-axes, respectively. The system is illustrated in Fig. 1.

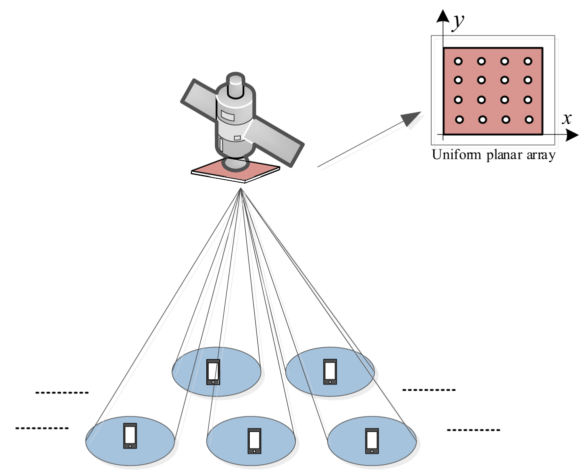

Fig. 1. Illustration of the LEO satellite downlink communication system.

A. LEO Satellite Downlink Channel Model

As different users are usually spatially separated by a certain distance, the channel realizations between the satellite and different users are assumed to be uncorrelated.

Thus, the complex baseband downlink space domain channel response between the LEO satellite and the k-th user, at instant t and frequency f, can be written as [15], [35]

$$\mathbf{h}_k = \mathrm{e}^{\left\{j2\pi\left[tv_k^{\mathrm{sat}}-f\tau_k^{\min}\right]\right\}}g_k(t,f)\mathbf{v}_k\in\mathbb{C}^{N_T\times 1}, \tag{1}$$

where v_k^{sat} represents the Doppler shifts caused by the motion of the LEO satellite, $\tau_k^{\min}$ is the minimum path delay of the k-th user, i.e., $\tau_k^{\min}=\min_l\{\tau_{k,l}\}$, where $\tau_{k,l}$ is the delay of the l-th path to user k. The downlink channel gain $g_k(t,f)$ of the k-th user is defined as

$$g_k(t,f)=\sum_{l=0}^{L_k-1}g_{k,l}e^{\left\{j2\pi\left[tv_{k,l}^{\mathrm{ut}}-f\tau_{k,l}^{\mathrm{ut}}\right]\right\}}, \tag{2}$$

where L_k is the number of propagation paths of the k-th user, $g_{k,l}$ is the complex-valued gain associated with path l of the k-th user, $v_{k,l}^{\mathrm{ut}}$ is the Doppler shifts for the path l caused by the motion of the k-th user, and $\tau_{k,l}^{\mathrm{ut}}\triangleq\tau_{k,l}-\tau_k^{\min}$. Noted that the LEO satellite communication systems are usually operated under line-of-sight (LOS) propagations, therefore, the Rician channel model can be used to describe the channel [15], [40]. Then, we assume the channel gain $g_k(t,f)$ follows Rician distribution with the Rician factor κ_k and power $\mathbb{E}\{|g_k(t,f)|^2\}=\sigma_g^2$. The UPA response vector $\mathbf{v}_k\in\mathbb{C}^{N_T\times 1}$ can be expressed by [41]

$$\mathbf{v}_k\left(\theta_k,\varphi_k\right)\triangleq\mathbf{v}_x\left(\theta_k,\varphi_k\right)\otimes\mathbf{v}_y\left(\theta_k\right), \tag{3}$$

where $\mathbf{v}_x(\theta_k,\varphi_k)\in\mathbb{C}^{M_x\times 1}$ and $\mathbf{v}_y(\theta_k)\in\mathbb{C}^{M_y\times 1}$ are the UPA response vectors of the angle with respect to the x- or y-axis defined by

$$\mathbf{v}_x\left(\theta_k,\varphi_k\right)=\frac{1}{\sqrt{M_x}}\Big[1,e^{-j\pi\sin(\theta_k)\cos(\varphi_k)},\ldots, e^{-j\pi(M_x-1)\sin(\theta_k)\cos(\varphi_k)}\Big]^T \tag{4}$$

and

$$\mathbf{v}_y\left(\theta_k\right)=\frac{1}{\sqrt{M_y}}\Big[1,e^{-j\pi\cos(\theta_k)},\ldots,e^{-j\pi(M_y-1)\cos(\theta_k)}\Big]^T \tag{5}$$

where φ_k and θ_k are the angles of the k-th user with respect to the x- and y-axes, respectively [15].

B. Downlink Transmission

Define s_k as the transmitted signal from the satellite to the k-th user, then the received signal of the k-th user can be written as

$$y_k=\mathbf{h}_k^H\mathbf{w}_ks_k+\mathbf{h}_k^H\sum_{i\neq k}\mathbf{w}_is_i+n_k, \tag{6}$$

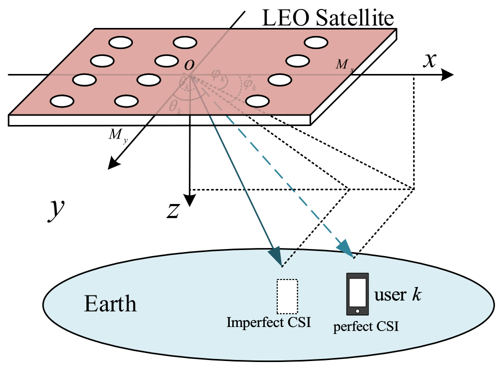

Fig. 2. The geometrical relation between the UPA and the k-th user with the angular mismatch.

where $\mathbf{w}_k \in \mathbb{C}^{N_T \times 1}$ represents the precoding vector of the k-th user, $n_k \sim \mathcal{CN}(0, \sigma_n^2)$ denotes the additive Gaussian white noise (AWGN) with zero mean and variance σ_n^2, and the power of the signal s_k is assumed to be 1, i.e., $\mathbb{E}\{|s_k|^2\} = 1$. The precoding vector $\mathbf{w}_k$ can be efficiently designed based on perfect AoD-based CSI according to different criteria, such as SINR balancing, sum-rate maximization, etc. However, in practical scenarios, due to the existing angle deviations of the LEO satellite positioning to the k-th user, it is difficult for the LEO satellite to obtain the perfect AoD-based CSI of each user. To reflect this inevitable angular uncertainty, we make some modifications to the channel response $\mathbf{h}_k$ in (1). Firstly, we construct the following relationship,

$$\varphi_k = \hat{\varphi}_k + \varphi_k^e + \bar{\varphi}, \tag{7}$$

$$\theta_k = \hat{\theta}_k + \theta_k^e + \bar{\theta}, \tag{8}$$

where $\hat{\varphi}_k$ and $\hat{\theta}_k$ represent the estimated angles of the k-th user obtained at the LEO satellite side, with respect to the x- or y-axis, respectively. In the practical systems, the estimated angles $\hat{\varphi}_k$ and $\hat{\theta}_k$ can be obtained based on the positioning system on satellites [18], or the calculation from the geographic position information between the k-th user and the satellite. Since $\hat{\varphi}_k$ and $\hat{\theta}_k$ are inaccurate angles for the k-th user, φ_k^e and θ_k^e are introduced to represent the angular deviations caused by the angle estimation error, where φ_k^e and θ_k^e can be modeled as random variables with a particular distribution (e.g., uniform distribution, Gaussian distribution, etc.), with respect to the perfect angle φ_k and θ_k, respectively. Furthermore, $\bar{\theta}$ and $\bar{\varphi}$ are used to represent the common angle deviation for all users caused by the attitude jitter of the satellite [16], [42]. Fig. 2 gives the illustration of the signal transmitting directions of the UPA with angular deviations for the k-th user.

Correspondingly, we can rewrite the UPA response vector $\mathbf{v}_k$ based on (7) and (8), i.e.,

$$\mathbf{v}_k \triangleq \mathbf{v}_x\left(\hat{\theta}_k + \theta_k^e + \bar{\theta}, \hat{\varphi}_k + \varphi_k^e + \bar{\varphi}\right) \otimes \mathbf{v}_y\left(\hat{\theta}_k + \theta_k^e + \bar{\theta}\right), \tag{9}$$

where $\mathbf{v}_x(\cdot)$ and $\mathbf{v}_y(\cdot)$ are defined in (4) and (5), respectively. The angle uncertainty is incorporated into the channel model defined in (1), reflected by the modified vector $\mathbf{v}_k$. Since the current satellite attitude estimation technologies can control the estimation error to 10^{-4} rad, and the actual influence of $\bar{\theta}$ and $\bar{\varphi}$ may be very small [16], so we omit $\bar{\theta}$ and $\bar{\varphi}$ in the following derivation. We will investigate the optimal precoding structure for robust precoder design in the following sections, assuming that the distribution information of φ_k^e and θ_k^e can be known.

C. Problem Formulation

Firstly, we define the k-th user's SINR_k as

$$\mathrm{SINR}_k = \frac{\mathbf{w}_k^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_k}{\sigma_n^2 + \sum_{i \neq k}^K \mathbf{w}_i^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_i}, \tag{10}$$

and the ergodic achievable rate $\mathcal{R}_k$ can be expressed as [6], [43]

$$\mathcal{R}_k = \mathbb{E}\left\{\log_2\left(1 + \frac{\mathbf{w}_k^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_k}{\sigma_n^2 + \sum_{i \neq k}^K \mathbb{E}\left\{\mathbf{w}_i^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_i\right\}}\right)\right\}, \tag{11}$$

where the interference-plus-noise $\mathbf{h}_k^H \sum_{i \neq k} \mathbf{w}_i s_i + n_k$ can be assumed as Gaussian noise, whose covariance is $\sigma_n^2 + \sum_{i \neq k}^K \mathbb{E}\{\mathbf{w}_i^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_i\}$. Because the ergodic rate $\mathcal{R}_k$ has no closed-form expression, it is hard to handle $\mathcal{R}_k$ directly, so we apply the following approximation according to the concavity of log function combined with Jensen's inequality [43],

$$\begin{aligned}\mathcal{R}_k \le \bar{\mathcal{R}}_k \triangleq \log_2\left(\sigma_n^2 + \mathbb{E}\left\{\sum_{i=1}^K \mathbf{w}_i^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_i\right\}\right) \\ -\log_2\left\{\sigma_n^2 + \mathbb{E}\left\{\sum_{i \neq k}^K \mathbf{w}_i^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_i\right\}\right\},\end{aligned} \tag{12}$$

where $\bar{\mathcal{R}}_k$ is the upper bound of $\mathcal{R}_k$. Note that the approximation in (12) is tight and has been verified to be theoretically accurate in [45]. In addition, we provide the simulation results in Section V to validate the numerical accuracy. To make the corresponding optimization tractable, we conduct the following optimization problem,

$$\begin{aligned}\mathbf{P1}: \underset{\{\mathbf{w}_k\}_{k=1}^K}{\text{maximize}} \quad & \sum_{k=1}^K \bar{\mathcal{R}}_k \\ \text{s.t.} \quad & \left[\sum_{k=1}^K \mathbf{w}_k \mathbf{w}_k^H\right]_{n,n} \le P_n, \forall k \in \mathcal{K}, \forall n \in \mathcal{N},\end{aligned} \tag{13}$$

where the upper bound $\bar{\mathcal{R}}_k$ is regarded as the optimization objective, P_n is the maximum available power for the n-th antenna. $\mathcal{K} = \{1, \ldots, K\}$ is the user index set and $\mathcal{N} = \{1, \ldots, N_T\}$ is the antenna index set.

III. Robust Precoder Design

Because the optimization problem (13) involves high-dimensional variables while the objective function is non-convex, it is intractable for an exact solution. Indeed, it is an NP-hard problem. In this section, we try to derive an efficient solution structure for this problem with certain transformation and simplification techniques.

A. Optimal Solution Structure

Suppose $\mathcal{R}_1^*, \mathcal{R}_2^*, \ldots, \mathcal{R}_K^*$ are the ergodic rates achieved by the optimal solutions of $\mathbf{P1}$, then $\mathbf{P1}$ can be equivalently transformed into the following QoS problem $\mathbf{P2}$ [3], [43],

$$\mathbf{P2}: \underset{\{\mathbf{w}_k\}_{k=1}^K}{\text{minimize}} \quad \sum_{k=1}^{K} \mathbf{w}_k^H \mathbf{w}_k$$
$$\text{s.t.} \begin{array}{l} \bar{\mathcal{R}}_k \geq \mathcal{R}_k^*, \forall k \in \mathcal{K}, \\ \left[\sum_{k=1}^{K} \mathbf{w}_k \mathbf{w}_k^H\right]_{n,n} \leq P_n, \forall k \in \mathcal{K}, \forall n \in \mathcal{N}. \end{array} \tag{14}$$

Compared with the global optimal solution of $\mathbf{P1}$, the solution of $\mathbf{P2}$ can achieve the same ergodic rates $\mathcal{R}_1^*, \mathcal{R}_2^*, \ldots, \mathcal{R}_K^*$ with the equal or even lower total power. In other words, the optimal solution of $\mathbf{P2}$ is not worse than that of $\mathbf{P1}$. The proof of the equivalence between the optimal solution of $\mathbf{P1}$ and $\mathbf{P2}$ is given in Appendix A.

By transforming $\mathbf{P1}$ into $\mathbf{P2}$, the ergodic rate $\bar{\mathcal{R}}_k$ of each user is decoupled to the constraint, which provides convenience for us to investigate the structure of the optimal robust precoder. The following equivalence can be performed,

$$\bar{\mathcal{R}}_k \geq \mathcal{R}_k^* \Leftrightarrow \mathbb{E}\left\{\text{SINR}_k\right\} \geq \gamma_k, \tag{15}$$

where $\gamma_k = 2^{\mathcal{R}_k^*} - 1$. Thus, $\mathbf{P2}$ can be rewritten as

$$\mathbf{P3}: \underset{\{\mathbf{w}_k\}_{k=1}^K}{\text{minimize}} \sum_{k=1}^{K} \mathbf{w}_k^H \mathbf{w}_k$$
$$\text{s.t.} \begin{array}{l} \mathbb{E}\left\{\text{SINR}_k\right\} \geq \gamma_k, \forall k \in \mathcal{K}, \\ \left[\sum_{k=1}^{K} \mathbf{w}_k \mathbf{w}_k^H\right]_{n,n} \leq P_n, \forall k \in \mathcal{K}, \forall n \in \mathcal{N}, \end{array} \tag{16}$$

where $\mathbb{E}\{\text{SINR}_k\}$ can be approximated as [11], [46]

$$\mathbb{E}\left\{\text{SINR}_k\right\} \approx \frac{\mathbf{w}_k^H \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \mathbf{w}_k}{\sigma_n^2 + \sum_{i \neq k}^{K} \mathbf{w}_i^H \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \mathbf{w}_i}. \tag{17}$$

Hence $\mathbb{E}\{\text{SINR}_k\} \geq \gamma_k$ can be written as

$$\mathcal{C}_k = \sum_{i \neq k} \frac{1}{\sigma_n^2} \mathbf{w}_i^H \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \mathbf{w}_i + 1 - \frac{1}{\gamma_k \sigma_n^2} \mathbf{w}_k^H \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \mathbf{w}_k \leq 0. \tag{18}$$

Note that strong duality and the KKT conditions of $\mathbf{P3}$ are necessary and sufficient for the optimal solution [3], [43]. Therefore, we first write the Lagrangian of $\mathbf{P3}$ as

$$\mathcal{L}_R = \sum_{k=1}^{K} \mathbf{w}_k^H \mathbf{w}_k + \sum_{k=1}^{K} v_k \mathcal{C}_k + \sum_{n=1}^{N_T} q_n \mathcal{P}_n, \tag{19}$$

where $\{v_k\}_{k=1}^K$ are the Lagrange multipliers for the QoS constraints, $\{q_n\}_{n=1}^{N_T}$ for the N_T PAPCs and $\mathcal{P}_n = (\mathbf{e}_n^T (\sum_{k=1}^K \mathbf{w}_k \mathbf{w}_k^H) \mathbf{e}_n - P_n)$, and the term $\sum_{n=1}^{N_T} q_n \mathcal{P}_n$ can be written in the following compact form,

$$\sum_{n=1}^{N_T} q_n \mathcal{P}_n = \operatorname{Tr}\left(\left(\sum_{k=1}^{K} \mathbf{w}_k \mathbf{w}_k^H - \mathbf{P}\right) \mathbf{Q}\right), \tag{20}$$

where the diagonal matrix $\mathbf{P} = \operatorname{diag}(P_1, P_2, \ldots, P_{N_T})$ and $\mathbf{Q} = \operatorname{diag}(q_1, q_2, \ldots, q_{N_T})$. The derivative of $\mathcal{L}_\mathcal{R}$ is given by

$$\frac{\partial \mathcal{L}_\mathcal{R}}{\partial \mathbf{w}_k} = \mathbf{w}_k + \sum_{i \neq k}^{K} \frac{v_i}{\sigma_n^2} \mathbb{E}\left\{\mathbf{h}_i \mathbf{h}_i^H\right\} \mathbf{w}_k - \frac{v_k}{\gamma_k \sigma_n^2} \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \mathbf{w}_k + \mathbf{Q} \mathbf{w}_k. \tag{21}$$

The optimal robust precoding vectors $\mathbf{w}_k^*$ of $\mathbf{P3}$ must satisfy the KKT conditions, i.e., $\frac{\partial \mathcal{L}_\mathcal{R}}{\partial \mathbf{w}_k} = 0, \forall k \in \mathcal{K}, \forall n \in \mathcal{N}$. Thus, we have

$$\frac{v_k}{\sigma_n^2} \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \mathbf{w}_k = \gamma_k \left(\mathbf{I} + \sum_{i \neq k}^{K} \frac{v_i}{\sigma_n^2} \mathbb{E}\left\{\mathbf{h}_i \mathbf{h}_i^H\right\} + \mathbf{Q}\right) \mathbf{w}_k. \tag{22}$$

We assume that $\mathbf{w}_k^* = \sqrt{\rho_k} \hat{\mathbf{w}}_k$, where ρ_k represents the precoding power and $\hat{\mathbf{w}}_k$ denotes the unit-norm precoding direction for the k-th user. Let $\mathbf{S}_k = \frac{v_k}{\sigma_n^2} \mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\}$ and $\mathbf{N}_k = \mathbf{I} + \sum_{i \neq k}^{K} \frac{v_i}{\sigma_n^2} \mathbb{E}\{\mathbf{h}_i \mathbf{h}_i^H\} + \mathbf{Q}$. By regarding (22) as a well-known generalized eigenvalue problem, we can obtain $\hat{\mathbf{w}}_k$ and γ_k by performing generalized eigenvalue decomposition on matrices $\mathbf{S}_k$ and $\mathbf{N}_k$, where γ_k corresponds to the maximum generalized eigenvalue of matrix pair $(\mathbf{S}_k, \mathbf{N}_k)$ and $\hat{\mathbf{w}}_k$ is the generalized eigenvector with respect to γ_k [43], i.e.,

$$\gamma_k = \text{max.generalized eigenvalue}\left(\mathbf{S}_k, \mathbf{N}_k\right), \tag{23}$$

$$\hat{\mathbf{w}}_k = \text{max.generalized eigenvector}\left(\mathbf{S}_k, \mathbf{N}_k\right). \tag{24}$$

It is worth noting that the constraint $\mathbb{E}\{\text{SINR}_k\} \geq \gamma_k$ in (16) holds with equality at the optimal solution of $\mathbf{P3}$, i.e.,

$$\sum_{i \neq k} \frac{1}{\sigma_n^2} \rho_i \hat{\mathbf{w}}_i^H \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \hat{\mathbf{w}}_i + 1 - \frac{1}{\gamma_k \sigma_n^2} \rho_k \hat{\mathbf{w}}_k^H \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \hat{\mathbf{w}}_k = 0. \tag{25}$$

Once $\hat{\mathbf{w}}_k$ and γ_k have been obtained, we can compute the precoding power ρ_k according to (25). Denote

$$[\mathbf{M}]_{k,i} = \begin{cases} \frac{1}{\gamma_i} \hat{\mathbf{w}}_i^H \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \hat{\mathbf{w}}_i, & k = i \\ -\hat{\mathbf{w}}_i^H \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} \hat{\mathbf{w}}_i, & k \neq i, \end{cases} \tag{26}$$

where $\mathbf{M} \in \mathbb{C}^{K \times K}$. Then, (25) can be rewritten as

$$\sum_{i=1}^{K} [\mathbf{M}]_{k,i} \rho_i = \sigma_n^2, k \in \mathcal{K}. \tag{27}$$

As a result, the precoding power vector $\boldsymbol{\rho} = [\rho_1, \ldots, \rho_K]^T$ can be computed by

$$\boldsymbol{\rho} = \sigma_n^2 \mathbf{M}^{-1} \mathbf{1}_{K \times 1}. \tag{28}$$

Consequently, the optimal robust precoding vectors can be computed through the solution structure, i.e., (24) and (28), with the Lagrange multipliers and the $\mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\}$ both determined. However, finding the Lagrange multipliers and deriving the closed-form expression of $\mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\}$ are not easy tasks. We will tackle these challenges in the following subsections.

B. Determine the Lagrange Multipliers q_n and v_k

To find the optimal Lagrange multipliers, we need to reveal the hidden relationship between q_n and v_k.

Firstly, we consider adopting the projected subgradient technique given in [19] to determine $\mathbf{Q}$. In the iterative expression, the projection operation is involved, i.e.,

$$\mathbf{Q}^{i+1} = \mathrm{proj}\left(\mathbf{Q}^i + t_i \mathrm{diag}\left(\mathrm{diag}\left(\sum_{k=1}^{K} \mathbf{w}_k \mathbf{w}_k^H\right) - \mathbf{p}\right)\right), \tag{29}$$

where $\mathbf{p} = [P_1, \ldots, P_{N_T}]^T$, i is the iteration index and t_i is the iteration step size with $t_i = 1/i$. According to the KKT conditions of (19): $q_n \geq 0, \forall n \in \mathcal{N}$ and $q_n \mathcal{P}_n = 0, \forall k \in \mathcal{K}, \forall n \in \mathcal{N}$, we can apply a more straightforward projection, the update equation for $\mathbf{Q}$ can be reformulated as

$$\mathbf{Q}^{i+1} = \max\left(\mathbf{Q}^i + t_i \mathrm{diag}\left(\mathrm{diag}\left(\sum_{k=1}^{K} \mathbf{w}_k \mathbf{w}_k^H\right) - \mathbf{p}\right), \mathbf{0}\right), \tag{30}$$

where the maximum operator is defined element-wise. The proof process is referred to [39]. By using (30), the optimal Lagrange multiplier $\hat{q}_n$ can be determined after the convergence of the iteration.

By observing the Lagrangian function given in (19) and the objective function of its dual problem, we can obtain the following relationship,

$$\sum_{k=1}^{K} v_k - \sum_{n=1}^{N_T} q_n P_n = \sum_{k=1}^{K} \mathbf{w}_k^H \mathbf{w}_k \leq P_{\mathrm{total}}, \tag{31}$$

where P_{total} is the maximum transmit power. (31) can be guaranteed by the strong duality between the problem **P3** and its dual problem [47], where the proof of the strong duality for the per-antenna power constraint problem is presented in [19]. Besides, by rearranged (22), we can obtain the fixed-point equations for v_k,

$$v_k^{-1} = \frac{1}{\sigma_n^2} \mathbb{E}\left\{\mathbf{h}_k^H \left(\mathbf{I} + \sum_{i=1}^{K} \frac{v_i}{\sigma_n^2} \mathbf{h}_i \mathbf{h}_i^H + \hat{\mathbf{Q}}\right)^{-1} \mathbf{h}_k\right\} \left(1 + \frac{1}{\gamma_k}\right), \tag{32}$$

where $\hat{\mathbf{Q}} = \mathrm{diag}(\hat{q}_1, \hat{q}_2, \ldots, \hat{q}_n)$. To maintain the equivalence of problems **P1** and **P3**, the SINR target γ_k in **P3** needs to be as large as possible under the given power constraints, while (31) and (32) must be satisfied. As a result, the following convex problem can be considered to determine v_k,

$$\begin{aligned} &\underset{v_k, \gamma_k}{\mathrm{maximize}} \ \textstyle\sum_{k=1}^{K} \log_2(1+\gamma_k) \\ &\mathrm{s.t.} \qquad (31) \quad \text{and} \quad (32). \end{aligned} \tag{33}$$

However, solving the above convex problem needs to perform high-dimensional matrix inversions, which increases the computational complexity to obtain v_k. Actually, the channels become more orthogonal as the number of antennas increases [48]. Considering the near-orthogonal characteristics of the channels, we can adopt the following approximation for each v_k [39],

$$v_k \approx \gamma_k \sigma_n^2 \left(\frac{1}{\beta_k} + \frac{\mathrm{Tr}\left(\hat{\mathbf{Q}} \mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\}\right)}{\beta_k^2}\right), \tag{34}$$

where $\beta_k = \|\mathbf{h}_k\|^2$. By replacing (32) with (34), we can calculate v_k efficiently by (33) using a water-filling algorithm.

To sum up, based on the iterative expression in (30) and the convex optimization problem (33), the Lagrange multipliers q_n and v_k can be easily determined. Besides, using the channel model given in (1) and (9), when φ_k^e and θ_k^e both exhibit a uniform distribution, i.e., $\varphi_k^e \sim U[\varphi_L, \varphi_U]$, $\theta_k^e \sim U[\theta_L, \theta_U]$, the element of m-th row and n-th column of matrix $\mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\}$ can be calculated as

$$\mathbb{E}\left\{\left[\mathbf{h}_k \mathbf{h}_k^H\right]_{m,n}\right\} = \begin{cases} \dfrac{\sigma_g^2}{M_x M_y}, \\ \quad n = m, a = b; \\ \dfrac{\sigma_g^2}{M_x M_y} \dfrac{1}{\Delta_\theta} \dfrac{2 e^{j\pi(n-m)\cos(\hat{\theta}_k)} \sin\left(\pi(n-m)\sin(\hat{\theta}_k)\theta_U\right)}{\pi(n-m)\sin(\hat{\theta}_k)}, \\ \quad n \neq m, a = b; \\ \dfrac{\sigma_g^2}{M_x M_y} \dfrac{4Z}{\Delta_\theta \Delta_\varphi} \dfrac{\sin\left(B\varphi_U \sin(\hat{\theta}_k)\right)\sin(\pi A \theta_U)}{\pi A B \sin(\hat{\theta}_k)}, \\ \quad n \neq m, a \neq b, \end{cases} \tag{35}$$

where

$$A = (b-a)\cos(\hat{\varphi}_k)\cos(\hat{\theta}_k) - (n - m - (b-a) M_y)\sin(\hat{\theta}_k), \tag{36}$$

$$B = \pi(b-a)\sin(\hat{\varphi}_k), \tag{37}$$

$$Z = e^{j\pi(n-m-(b-a)M_y)\cos(\hat{\theta}_k) + j\pi(b-a)\cos(\hat{\varphi}_k)\sin(\hat{\theta}_k)}. \tag{38}$$

The calculation process is provided in Appendix B-A.

Similarly, when φ_k^e and θ_k^e both exhibit a Gaussian distribution, i.e., $\theta_k^e \sim \mathcal{N}(\mu_{\theta,k}, \sigma_{\theta,k}^2)$, $\varphi_k^e \sim \mathcal{N}(\mu_{\varphi,k}, \sigma_{\varphi,k}^2)$, we can obtain

$$\mathbb{E}\left\{\left[\mathbf{h}_k \mathbf{h}_k^H\right]_{m,n}\right\} = \begin{cases} \dfrac{\sigma_g^2}{M_x M_y}, & n = m, a = b; \\ \dfrac{\sigma_g^2}{M_x M_y} e^{jQ - jP\mu_{\theta,k} - \frac{1}{2}P^2 \sigma_{\theta,k}^2}, & n \neq m, a = b; \\ \dfrac{\sigma_g^2}{M_x M_y} e^{\left(\frac{1}{2}C^2 \sigma_{\theta,k}^2 + C\mu_{\theta,k}\right) + E}, & n \neq m, a \neq b, \end{cases} \tag{39}$$

where

$$P = \pi(n-m)\sin(\hat{\theta}_k), \tag{40}$$

$$Q = \pi(n-m)\cos(\hat{\theta}_k), \tag{41}$$

$$D = \pi(b-a)\cos(\hat{\varphi}_k) - \pi(b-a)\sin(\hat{\varphi}_k)\mu_{\varphi,k}, \tag{42}$$

$$F = \pi^2 \sigma_{\varphi,k}^2 ((b-a)\sin(\hat{\varphi}_k))^2, \tag{43}$$

$$\begin{aligned} C = {} & jD\cos(\hat{\theta}_k) - \frac{1}{2} F \sin(2\hat{\theta}_k) \\ & - j\pi(n - m - (b-a) M_y)\sin(\hat{\theta}_k), \end{aligned} \tag{44}$$

$$E = jD\sin(\hat{\theta}_k) - \frac{1}{4} F\left(1 - \cos(2\hat{\theta}_k)\right)$$

Algorithm 1: Robust Precoder Design With Imperfect AoD-Based CSI Under PAPCs

1: Initialize $\mathbf{Q}^0 = \mathbf{I}$ and the estimated AoD angles $(\hat{\varphi}_k, \hat{\theta}_k)$ for k-th user. Give the parameters $(\varphi_L, \varphi_U, \theta_L, \theta_U)$ or $(\mu_{\varphi,k}, \sigma^2_{\varphi,k}, \mu_{\theta,k}, \sigma^2_{\theta,k})$. Set $i = 0$, P_n and ε_n,
2: Calculate $\mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\}$ using (35) or (39).
3: **while** $[\sum_{k=1}^{K} \mathbf{w}_k \mathbf{w}_k^H]_{n,n} - P_n > \varepsilon_n$ for any n **do**
4: Find v_k by solving (33).
5: Obtain γ_k using (23) and $\hat{\mathbf{w}}_k$ using (24).
6: Find the optimal power $\boldsymbol{\rho}$ allocated to K user using (28).
7: Calculate the precoding vector $\mathbf{w}_k^* = \sqrt{\rho_k}\hat{\mathbf{w}}_k$.
8: Update $\mathbf{Q}^{i+1}$ using (30) and the iteration step size t_i with $t_i = 1/i$.
9: $i = i + 1$.
10: **end while**
11: **return**the optimal precoder $\mathbf{w}_k^*$.

$$+ j\pi\left(n - m - (b-a)M_y\right)\cos(\hat{\theta}_k). \tag{45}$$

The calculation process for the case of the Gaussian distribution is presented in Appendix B-B.

Following the above development, we summarize the proposed robust precoder design algorithm in Algorithm 1. A simple termination strategy is applied here to stop the iteration process, i.e., $[\sum_{k=1}^{K} \mathbf{w}_k \mathbf{w}_k^H]_{n,n} - P_n > \varepsilon_n, \forall n \in \mathcal{N}$, where ε_n is the maximum allowable violation of the power constraint for the n-th antenna. Note that the dual problem of **P3** can be considered as a function of the Lagrange multiplier $\mathbf{Q}$, i.e. $f(\mathbf{Q})$. Therefore, based on the *Proposition 3* in [19], the subgradient projection method defined by (30) is guaranteed to converge to the global optimum of $f(\mathbf{Q})$ due to the concavity of $f(\mathbf{Q})$, which also guarantees the convergence of the proposed Algorithm 1.

IV. Low Complexity Implementation Based on Deep Learning

Note that Algorithm 1 needs to be iterated many times until each antenna meets its power constraints. In each iteration, the Lagrangian multipliers v_k and q_n, along with a complex generalized eigenvalue problem, need to be calculated, which leads to the high computational complexity. In the high-speed LEO satellite communication scenarios, Algorithm 1 may not be able to meet the demand for real-time processing. Since the closed-form solutions of the Lagrange multipliers is unavailable, the optimal parameters are obtained until iterative convergence. To avoid iteration and further reduce the computational complexity, the Lagrangian multipliers should be predicted directly. If the model-driven method is used for prediction, the accuracy and generalization of the model need to be optimized simultaneously. Such a method will inevitably lead to prediction error caused by the high dimension of the parameter matrix [29]. However, the DL-based method can directly learn the relationship between the samples and results [27], [28], [49], [50]. By improving the quality of the data set, the accuracy and generalization of the model can be considered simultaneously, and the complexity can be effectively controlled by adjusting the structure and size of the DL neural network. Therefore, we next apply the DL techniques for the low-complexity implementation of our proposed precoding approach.

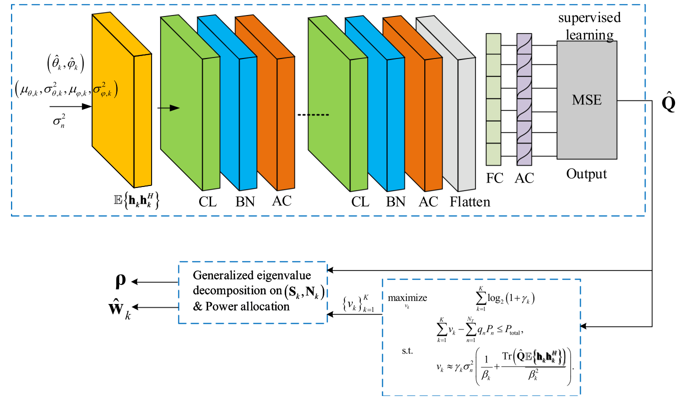

Fig. 3. The structure of the DL network for predicting the optimal Lagrange multipliers $\hat{\mathbf{Q}}$ for the case of Gaussian distribution, followed by the calculation process of the optimal precoding vectors.

In general, it is very challenging to learn the precoding vectors $\mathbf{w}_k$ directly. Note that once the optimal Lagrangian multipliers $\hat{\mathbf{Q}}$ are obtained, $\{v_k\}_{k=1}^K$ and the optimal precoding vector $\mathbf{w}_k^*$ can be obtained without iteration. Because of the large dimension of the channel matrix and the complex relationship between the Lagrangian multipliers $\mathbf{Q}$ and the channel information, the used DL neural network should have good feature simplification, generalization, and nonlinear calculation ability. However, there are many neural networks with powerful feature extraction and approximation capabilities [24]–[26]. Without loss of generality, we choose the convolutional neural network (CNN) as the backbone to predict $\mathbf{Q}$ through the channel matrices [29], [49], [50]. It is worth noting that this section aims to present a generic method using neural networks to reduce the computational complexity of the proposed precoding algorithm. Therefore, more in-depth research on the training and optimization of neural network structures for better performance can be left to future work.

A. Structure of the Neural Network for Predicting Lagrange Multipliers

As illustrated in Fig. 3, the DL network is composed of an input layer, several convolutional layer (CL) groups, a fully-connected (FC) layer group and an output layer. Each CL group is composed of three functional layers, including a CL, a batch normalization (BN) layer, and an activation (AC) layer. The FC layer group consists of a flatten layer, several hidden FC layers, an AC layer and an output layer [30], [43].The detailed description and modeling of the DL network are given in Appendix C. The goal of the DL network is to predict the optimal Lagrange multipliers $\hat{\mathbf{Q}}$ from the channel autocorrelation matrices $\mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\}$. According to the models given in (35) and (39), we first denote the channel matrix $\mathbf{H}_{ori}$ as

$$\mathbf{H}_{ori} = \left[\mathbb{E}\left\{\mathbf{h}_1 \mathbf{h}_1^H\right\}, \ldots, \mathbb{E}\left\{\mathbf{h}_K \mathbf{h}_K^H\right\}\right] \in \mathbb{C}^{N_T \times N_T \times K}, \tag{46}$$

$$\mathbf{H} = \left[\mathrm{Re}(\mathbf{H}_{ori}), \mathrm{Im}(\mathbf{H}_{ori})\right] \in \mathbb{C}^{N_T \times N_T \times 2K}. \tag{47}$$

Since current neural networks cannot operate on complex channel coefficients directly, data transformations are required before the input layer. One commonly-used way is to separate the complex channel coefficients into in-phase component $\text{Re}(\mathbf{H}_{ori})$ and quadrature component $\text{Im}(\mathbf{H}_{ori})$, i.e., (47). Besides, the predicted Lagrange multipliers are positive real values so that the sigmoid function can be chosen as the activation function in the activation layer [30],

$$\text{sigmoid}(x) = \frac{1}{1+e^{-x}}, \tag{48}$$

which increases the nonlinearity of the network.

The Lagrange multiplier matrix $\mathbf{Q}$ is not only related to the channel autocorrelation matrices but also related to the signal-to-noise ratio (SNR) at the precoder. For convenience, we involve SNR into the channel coefficients matrices, and then the training samples input to the neural network are represented as

$$\bar{\mathbf{H}} = \frac{\mathbf{H}}{\text{SNR}}, \tag{49}$$

where $\text{SNR} = P_T/\sigma_n^2$, P_T is the transmit power of the system, (49) is calculated element-wise, i.e., the (m,n,k)-th element of $\bar{\mathbf{H}}$ is defined as $(\bar{\mathbf{H}})_{m,n,k} = (\mathbf{H})_{m,n,k}/\text{SNR}$. Thus, we can describe the function of the used DL network as

$$\hat{\mathbf{Q}} = f_{\mathbf{Q}}\left(\bar{\mathbf{H}}\right). \tag{50}$$

In particular, by observing the expression given in (35) or (39), we can note that the channel matrix $\bar{\mathbf{H}}$ is only related to the estimated angle information $(\hat{\varphi}_k, \hat{\theta}_k)$, the parameters $(\varphi_L, \varphi_U, \theta_L, \theta_U)$ for the case of uniform angular error distribution or $(\mu_{\varphi,k}, \sigma_{\varphi,k}^2, \mu_{\theta,k}, \sigma_{\theta,k}^2)$ for Gaussian distribution, and the noise variance σ_n^2. For convenience, we define a new sample set $\mathcal{I}$ as (51) for the case of uniform distribution and (52) for Gaussian distribution.

$$\mathcal{I} = \left\{\left(\hat{\varphi}_k, \hat{\theta}_k\right), \left(\varphi_L, \varphi_U, \theta_L, \theta_U\right), \sigma_n^2\right\}, \forall k \in \mathcal{K}, \tag{51}$$

$$\mathcal{I} = \left\{\left(\hat{\varphi}_k, \hat{\theta}_k\right), \left(\mu_{\varphi,k}, \sigma_{\varphi,k}^2, \mu_{\theta,k}, \sigma_{\theta,k}^2\right), \sigma_n^2\right\}, \forall k \in \mathcal{K}. \tag{52}$$

Now, the structure of $\mathbb{E}\{\mathbf{h}_k\mathbf{h}_k^H\}$ given in (35) or (39) can be involved in the proposed DL network, instead of inputting $\bar{\mathbf{H}}$.

As a result, the function of the trained neural network can be redefined as

$$\hat{\mathbf{Q}} = f_{\mathbf{Q}}\left(\mathcal{I}\right), \tag{53}$$

where the function f describes the neural network structure to predict $\hat{\mathbf{Q}}$ based on the channel parameters.

B. Dataset Generation

Based on a sample set $\mathcal{I}$, the proposed Algorithm 1 can be used to calculate the corresponding Lagrange multipliers $\mathbf{Q}$ as the training labels. Since $\mathbf{Q}$ is a diagonal matrix, we let the column vector $\mathbf{q} = \text{diag}(\mathbf{Q})$ as the new training labels. The process of dataset generation is given in Algorithm 2. To ensure that the generated dataset is close to the real satellite communication scenarios, the parameter configuration for Algorithm 2 is referred to the typical LEO satellite constellations (such as SpaceX's "Starlink") and their Federal Communications Commission (FCC) filings [1], [44]. In addition, aimed at generating enough and comprehensive samples, we randomly generate the noise variance σ_n^2 as well as the user locations within the satellite beam coverage, where the estimated angles $\hat{\phi}_k$ and $\hat{\theta}_k$ are both set in the reasonable range $[20°, 160°]$.

Algorithm 2: Dataset Generation for Deep Learning.

1: **input**: The number of samples $N_{\mathcal{S}}$.
2: Set $i = 1$.
3: **while** $i < N_{\mathcal{S}}$ **do**
4: Generate the set $\mathcal{I}^{(i)}$ and the channel autocorrelation matrices $\mathbf{H}^{(i)}$ using (35) or (39). Generate the samples $\bar{\mathbf{H}}^{(i)}$ using (49).
5: Use the proposed iterative Algorithm 1 to solve problem (13), generating the corresponding training labels $\mathbf{Q}^{(i)}$.
6: Let $\mathbf{q}^{(i)} = \text{diag}(\mathbf{Q}^{(i)})$.
7: Put the i-th sample $\{\mathcal{I}^{(i)}, \bar{\mathbf{H}}^{(i)}, \mathbf{q}^{(i)}\}$ into the set $\mathcal{S}$.
8: Set $i = i + 1$.
9: **end while**
10: **return** The sample set $\mathcal{S}$.

Apart from the functional layers given in Fig. 3, we choose the mean squared error (MSE) as the loss function in our neural network, which is defined by

$$\text{MSE} = \frac{1}{N_{\mathcal{S}}}\sum_{i=1}^{N_{\mathcal{S}}}\left\|\mathbf{q}^{(i)} - \hat{\mathbf{q}}^{(i)}\right\|_2^2, \tag{54}$$

where $\hat{\mathbf{q}}^{(i)}$ is the the i-th predicted results in the neural network and $\mathbf{q}^{(i)}$ is the target results [30]. After all, the training can be conducted with typical Adam optimizer [51].

V. Simulation Results

Simulation results are presented to verify the effectiveness of the proposed robust precoding approach in the satellite downlink channel. Under different angular deviations, we evaluate the sum-rate performance of the proposed robust precoding. To show the superiority of the proposed schemes, we also simulate the RZF solution under PAPC given in [49] and the SLNR solution under PAPC in [38] for comparison. Besides, the sum-rate performance of different precoding approaches for both perfect and imperfect CSI is compared. Furthermore, a neural network is trained based on the dataset generated by Algorithm 2 to implement a low-complexity version of our precoding approach. As an example, we only train the neural network for the case that both φ_k^e and θ_k^e exhibit Gaussian distributions, where (39) is used to generate the i-th sample $\mathbf{H}^{(i)}$ with zero mean and random variance. The achievable sum rate of the LEO satellite systems is calculated by

$$\mathcal{R}_k = \log_2\left(1 + \frac{(\mathbf{w}_k^*)^H\mathbf{h}_k\mathbf{h}_k^H\mathbf{w}_k^*}{\sigma_n^2 + \sum_{i\neq k}^{K}(\mathbf{w}_i^*)\mathbf{h}_k\mathbf{h}_k^H\mathbf{w}_i^*}\right), \tag{55}$$

TABLE I
SIMULATION SETUP PARAMETERS

Parameters	Values
Frequency f	2 GHz
Height	600 Km
Number of antennas N_T	128
Number of antennas on x- and y-axes	8×16
Number of service users K	32
Maximum per antenna power P_n	2 W
Bandwidth B	20 MHz
Minimum Elevation Angle of each user	20°
Estimated angles $\hat{\varphi}_k$ for K users	$20^\circ \leq \hat{\varphi}_k \leq 160^\circ$
Estimated angles $\hat{\theta}_k$ for K users	$20^\circ \leq \hat{\theta}_k \leq 160^\circ$
Angular deviation $\varphi_k^e = \varphi^e$	$-5^\circ \sim 5^\circ$
Angular deviation $\theta_k^e = \theta^e$	$-5^\circ \sim 5^\circ$
Uniform distribution $[\varphi_L/\theta_L, \varphi_U/\theta_U]$	$[-5^\circ, 5^\circ]$ and $[-8^\circ, 8^\circ]$
Gaussian distribution (μ, σ)	$(0^\circ, 3^\circ)$ and $(0^\circ, 7.5^\circ)$

where $\mathbf{w}_k^*$ are the obtained optimal precoding vectors based on the perfect CSI, imperfect CSI or the expression $\mathbb{E}\{\mathbf{h}_k\mathbf{h}_k^H\}$. Substituting the channel model (1) into (55), we have

$$\mathcal{R}_k = \log_2\left(1 + \frac{|g_k(t,f)|^2 (\mathbf{w}_k^*)^H \mathbf{v}_k \mathbf{v}_k^H \mathbf{w}_k^*}{\sigma_n^2 + |g_k(t,f)|^2 \sum_{i\neq k}^K (\mathbf{w}_i^*) \mathbf{v}_k \mathbf{v}_k^H \mathbf{w}_i^*}\right). \quad (56)$$

It can be observed that the expression $\mathcal{R}_k$ has no direct relationship with the Doppler shifts v_k^{sat} and the minimum propagation delays $\tau_k^{\min}$, which are reflected in the expression of $g_k^{dl}(t,f)$. Thus, we omit the specific values for v_k^{sat} and $\tau_k^{\min}$.

We define G/T as the power gain G to equivalent noise temperature T of the receive antenna at the terminal, measured in dB/K, where K refers to degree Kelvin. The ratio G/T can describe the "quality" of an earth terminal, which is usually referred to the *figure of merit* of the receiving terminal [52]. According to the link budget analysis of the satellite downlink [1], the relationship between G/T and receiving signal-to-noise ratio SNR can be given by

$$\frac{G}{T} = \text{SNR} - \text{EIRP}_{sat} + L_{free} + L_{other} + 10\log_{10}(k_{Bol}B) \ \text{(dB/K)}, \quad (57)$$

where EIRP_{sat} is the effective radiated power referenced to an isotropic source (EIRP) of the LEO satellite antenna, L_{free} represents the free space path loss, L_{other} denotes the transmission loss (i.e., atmospheric loss, rain loss, etc.), $k_{Bol} = 1.38 \times 10^{-23}\text{W}/(\text{Hz}\cdot\text{K})$ is the Boltzmann constant, and B is the system bandwidth. Aimed at revealing the performance of the proposed downlink precoder under different terminals, we use G/T as the x-axis in the simulations. Simulation parameters are extracted from [53], and given in Table I.

The convergence performance of Algorithm 1 is first presented in Fig. 4. We use the violation probability to describe the proportion of antennas transmitting a power that is more than 10% higher than P_n among N_T PAPCs, i.e., $\varepsilon_n = 0.1P_n$. It can be observed from Fig. 4 that within some iterations, the PAPCs can be met in most cases.

Fig. 5 gives the comparison between the exact average sum rate and the approximate sum rate, where the exact system

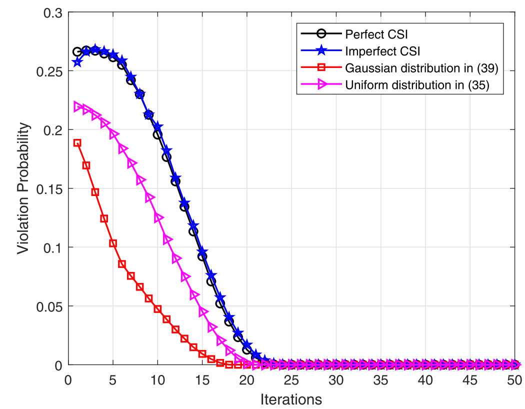

Fig. 4. Iteration performance of Algorithm 1. The violation probability measures the proportion of antennas transmitting a power that is more than 10% higher than P_n among N_T PAPCs, in 1,000 channel realizations. (Set $\varepsilon_n = 0.1P_n$).

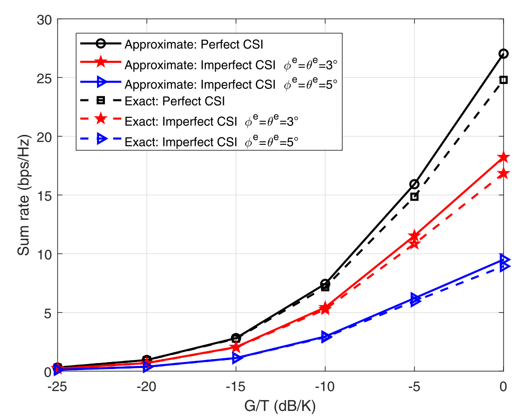

Fig. 5. Comparison between the exact average sum rate and the approximate sum rate.

average sum rate $\mathcal{R}_{sum}$ is calculated by

$$\mathcal{R}_{sum} = \sum_{k=1}^K \mathcal{R}_k = \mathbb{E}\left\{\log_2\left(1 + \frac{\mathbf{w}_k^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_k}{\sigma_n^2 + \sum_{i\neq k}^K \mathbb{E}\left\{\mathbf{w}_i^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_i\right\}}\right)\right\}, \quad (58)$$

and the approximate sum rate $\bar{\mathcal{R}}_{sum}$ is equal to

$$\bar{\mathcal{R}}_{sum} = \sum_{k=1}^K \bar{\mathcal{R}}_k = \log_2\left(1 + \frac{\mathbb{E}\left\{\mathbf{w}_k^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_k\right\}}{\sigma_n^2 + \sum_{i\neq k}^K \mathbb{E}\left\{\mathbf{w}_i^H \mathbf{h}_k \mathbf{h}_k^H \mathbf{w}_i\right\}}\right). \quad (59)$$

It can be seen that for different G/Ts, the residuals between the exact sum rate $\mathcal{R}_{sum}$ and the approximate sum rate $\bar{\mathcal{R}}_{sum}$ are small enough to ignore, verifying the rationality of (12).

Fig. 6, Fig. 7 and Fig. 8 show the sum-rate performance of the proposed Algorithm 1 versus different G/Ts. For convenience, we take $\varphi_k^e = \varphi^e = 3^\circ$ and $\theta_k^e = \theta^e = 3^\circ$ as the first imperfect CSI observation sample, and $\varphi_k^e = \varphi^e = 5^\circ$ and $\theta_k^e = \theta^e = 5^\circ$ as the second imperfect CSI sample for all users.

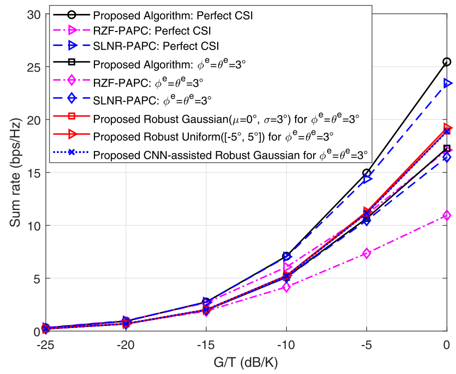

Fig. 6. Sum-rate performance of Algorithm 1 with the imperfect AoD-based CSI. Suppose the angular deviations φ_k^e and θ_k^e both exhibit a uniform distribution in $[-5^\circ, 5^\circ]$, or exhibit a Gaussian distribution with mean $\mu = 0^\circ$ and the standard deviation $\sigma = 3^\circ$. Take $\varphi^e = \theta^e = 3^\circ$ as the imperfect CSI observation sample.

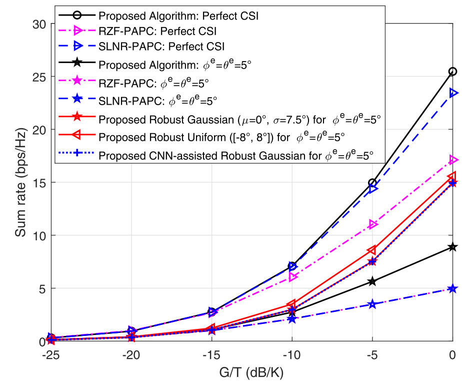

Fig. 8. Sum-rate performance of Algorithm 1 with the imperfect AoD-based CSI. Suppose the angular deviations φ_k^e and θ_k^e both exhibit a uniform distribution in $[-8^\circ, 8^\circ]$, or exhibit a Gaussian distribution with mean $\mu = 0^\circ$ and the standard deviation $\sigma = 7.5^\circ$. Take $\varphi^e = \theta^e = 5^\circ$ as the imperfect CSI observation sample.

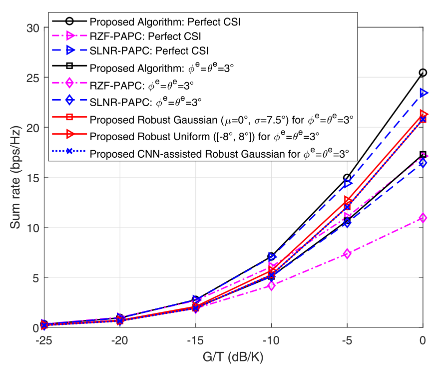

Fig. 7. Sum-rate performance of Algorithm 1 with the imperfect AoD-based CSI. Suppose the angular deviations φ_k^e and θ_k^e both exhibit a uniform distribution in $[-8^\circ, 8^\circ]$, or exhibit a Gaussian distribution with mean $\mu = 0^\circ$ and the standard deviation $\sigma = 7.5^\circ$. Take $\varphi^e = \theta^e = 3^\circ$ as the imperfect CSI observation sample.

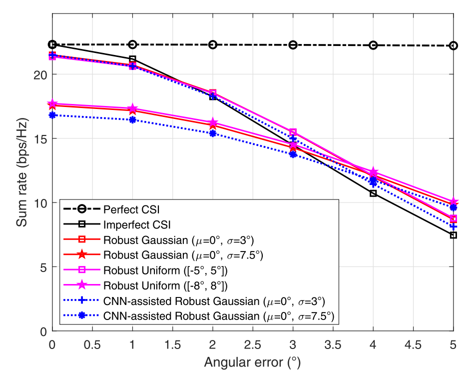

Fig. 9. Sum-rate performance of Algorithm 1 versus the angular deviations θ_k^e and ϕ_k^e from 0° to 5° with $\mathrm{G/T} = 0$ dB/K.

The notation 'Proposed Robust Gaussian $(\mu = 0^\circ, \sigma = 3^\circ)$ for $\varphi^e = \theta^e = 3^\circ$' or 'Proposed Robust Gaussian $(\mu = 0^\circ, \sigma = 7.5^\circ)$ for $\varphi^e = \theta^e = 3^\circ/5^\circ$' represents the sum-rate performance when both φ_k^e and θ_k^e exhibit a Gaussian distribution with mean $\mu = 0^\circ$ and the standard deviation $\sigma = 3^\circ$ or 7.5° with respect to the case '$\varphi^e = \theta^e = 3^\circ$' and '$\varphi^e = \theta^e = 5^\circ$,' where '$\varphi^e = \theta^e = 3^\circ$' and '$\varphi^e = \theta^e = 5^\circ$' can be regarded as a baseline in these cases. Similarly, the notation 'Proposed Robust Uniform $([-5^\circ, 5^\circ])$ for $\varphi^e = \theta^e = 3^\circ$' or 'Proposed Robust Uniform $([-8^\circ, 8^\circ])$ for $\varphi^e = \theta^e = 3^\circ/5^\circ$' represents the sum-rate performance when both φ_k^e and θ_k^e exhibit a uniform distribution in $[-5^\circ, 5^\circ]$ or $[-8^\circ, 8^\circ]$ with respect to '$\varphi^e = \theta^e = 3^\circ$' and '$\varphi^e = \theta^e = 5^\circ$,' respectively. In Fig. 6, Fig. 7 and Fig. 8, we can see that the proposed precoding scheme can achieve better sum-rate performance compared with the RZF solution and the SLNR solution under PAPC for both perfect and imperfect CSI (including '$\varphi^e = \theta^e = 3^\circ$' and '$\varphi^e = \theta^e = 5^\circ$'). In particular, when $\varphi^e = \theta^e = 5^\circ$, the case 'Proposed Robust Gaussian $(\mu = 0^\circ, \sigma = 7.5^\circ)$' can achieve about 100% sum-rate performance gain compared with the case 'RZF-PAPC' and 'SLNR-PAPC'.

Compared with the cases in Fig. 6, Fig. 7 shows that the sum-rate performance of the robust precoding scheme can realize a much better performance for both uniform and Gaussian distribution by adjusting the distribution parameters of the angle deviations. Moreover, as can be seen from Fig. 8, the sum-rate performance of the proposed precoding scheme decreases as the angle deviation φ_k^e and θ_k^e increases. Especially in the case '$\varphi^e = \theta^e = 5^\circ$,' there exists nearly 60% loss in the sum-rate performance compared with the case 'Perfect CSI'. Based on the derived expression $\mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\}$, the proposed robust precoding scheme can achieve a significant sum-rate performance gain compared with the case '$\varphi^e = \theta^e = 5^\circ$'. Furthermore, the performance gains are small at low G/T, but become significant as G/T increases.

In addition, we further measure the sum-rate performance versus different angular deviations, and the results are presented in Fig. 9. It can be seen that when the angular errors are assumed to exhibit the Gaussian distribution or uniform distribution in different ranges, the robustness of the proposed precoding algorithm under different angle errors is different. In particular, compared with "Robust Gaussian $(\mu = 0^\circ, \sigma = 7.5^\circ)$," small angular deviations are more likely to occur in the case of "Robust Gaussian $(\mu = 0^\circ, \sigma = 3^\circ)$". Hence, the sum-rate performance

TABLE II
COMPARISON OF COMPLEXITY BETWEEN ALGORITHM 1 AND ITS LOW COMPLEXITY VERSION WITH CNN

Algorithm	Complexity	Typical Value	Computation Time
Algorithm 1	$O\left(N_{\text{iter}}\left(K^2(N_T+K+1)+(KN_T+N_T^3)(N_T+1)\right)\right)^{\text{a}}$	8.1368E9	15.3732 s
Low-complexity with CNN	$O\left(N_T^2\left(2K\sum_{c\in\mathcal{C},c\neq 1}k_c^h k_c^w k_c^l n_c n_{c-1}+\sum_{h\in\mathcal{H},h\neq 1}2^h+1\right)\right)^{\text{b}}$	3.6241E9	6.5670 s

[a] N_{iter} is the number of iterations required by Algorithm 1.
[b] Appendix C gives the meaning of the symbol notations.

of "Robust Gaussian ($\mu=0^\circ,\sigma=3^\circ$)" is better than that of "Robust Gaussian ($\mu=0^\circ,\sigma=7.5^\circ$)" at a small angular deviation. Similarly, the probability of the large angular deviations in "Robust Gaussian ($\mu=0^\circ,\sigma=7.5^\circ$)" is higher than that in "Robust Gaussian ($\mu=0^\circ,\sigma=3^\circ$)," resulting in that the sum-rate performance of "Robust Gaussian ($\mu=0^\circ,\sigma=7.5^\circ$)" is better than that of "Robust Gaussian ($\mu=0^\circ,\sigma=3^\circ$)" at a large angular deviation. Therefore, the sum rate downgrade with different speeds as the angular deviations increase. It can be concluded that our robust precoding approach can achieve a satisfactory performance gain by adjusting appropriate parameters $[\varphi_L,\varphi_U]$ and $[\theta_L,\theta_U]$ for the case of the uniform distribution, or $(\mu_{\theta,k},\sigma^2_{\theta,k})$ and $(\mu_{\phi,k},\sigma^2_{\phi,k})$ for the Gaussian distribution based on the expressions of $\mathbb{E}\{\mathbf{h}_k\mathbf{h}_k^H\}$.

As for the DL-based robust precoding approach, the Lagrangian multipliers can be directly obtained by applying the offline trained neural network so that the iterative process in Algorithm 1 is skipped and the optimal precoding vectors $\mathbf{w}_k^*$ can be directly calculated. Based on the DL network model in Appendix C, the complexity of DL-based robust precoding approach is summarized in Table II together with Algorithm 1. Under the same operating environment and configuration, it can be seen that the DL-based robust precoding approach reduces the computation time by 57.3% compared with Algorithm 1. Besides, in Fig. 6–Fig. 9, we use the notation 'Proposed CNN-assisted Robust Gaussian' or 'CNN-assisted Robust Gaussian' to represent the sum-rate performance of the proposed robust precoding based on the trained CNN, and the simulation results all show that the DL-based robust precoding approach can achieve satisfying sum-rate performance with lower computational complexity. In summary, the proposed robust precoding method can be implemented in a low-complexity form with the help of the trained neural networks and realize good robustness to the CSI with the angular mismatch in the AoD angles. The numerical results verify the correctness and effectiveness of the proposed robust approaches.

VI. CONCLUSION

In this paper, we investigated the robust precoder design for the LEO satellite downlink. We first developed an angular uncertainty-based channel model to describe the angular mismatch in the AoD of each user. Based on this model, we explored the optimal precoding structure to maximize the ergodic sum rate while satisfying PAPCs. By transforming the sum-rate maximization problem into an improved QoS problem, the optimal solution structure was obtained, where the precoding directions and powers can be calculated according to the solution of a generalized eigenvalue problem. We then developed an iterative algorithm to determine the Lagrange multipliers and further obtained the optimal precoding vectors. Aimed at reducing computational complexity and meeting the demand for real-time applications, a neural network was designed to learn the mapping relationship from CSI to the Lagrange multipliers to avoid iterations in the proposed algorithm. Simulation results demonstrated the effectiveness of the proposed robust precoding approach, which achieved a satisfactory balance between the performance and the computational complexity.

APPENDIX A
PROOF OF THE EQUIVALENCE BETWEEN **P1** AND **P2**

The proof of the equivalence between the optimal solution of **P1** and **P2** can be guaranteed by two steps: the first step is the equivalence of the solution of **P1** and **P2**; the second step is that strong duality and KKT conditions guarantee the optimality of the solution structure of **P2** problem. The proof process is given as follows:

1) Denote $\mathbf{w}_k=\sqrt{\rho_k}\bar{\mathbf{w}}_k$, where ρ_k is the power allocated to the k-th user, $\bar{\mathbf{w}}_k$ is the normalized precoding vector satisfying $\bar{\mathbf{w}}_k^H\bar{\mathbf{w}}_k=1$. Then the objective function of **P2** can be rewritten as

$$\sum_{k=1}^{K}\mathbf{w}_k^H\mathbf{w}_k=\sum_{k=1}^{K}\rho_k. \tag{60}$$

Define the optimal solution and corresponding ergodic rates of **P2** as $(\rho_1^\star,\ldots,\rho_K^\star,\bar{\mathbf{w}}_1^\star,\ldots,\bar{\mathbf{w}}_K^\star)$ and $\bar{\mathcal{R}}_1^\star,\ldots,\bar{\mathcal{R}}_K^\star$, respectively. Because of the constraint $\bar{\mathcal{R}}_k\geq\mathcal{R}_k^*$, we can assume there exists $\bar{\mathcal{R}}_m^\star$ satisfying

$$\bar{\mathcal{R}}_m^\star>\mathcal{R}_m^*. \tag{61}$$

It is easy to verify that $\bar{\mathcal{R}}_k$ monotonically increases with the power allocated to itself ρ_k and decreases with the power allocated to other user ρ_i, $i\neq k$. As $\bar{\mathcal{R}}_k$ is continuous with respect to ρ_m, there always exists a sufficiently small ε to establish a solution $(\rho_1^\star,\ldots,\rho_m^\star-\varepsilon,\ldots,\rho_K^\star,\bar{\mathbf{w}}_1^\star,\ldots,\bar{\mathbf{w}}_K^\star)$ with the corresponding rates $\hat{\mathcal{R}}_1,\ldots,\hat{\mathcal{R}}_K$ satisfying

$$\hat{\mathcal{R}}_k=\begin{cases}\bar{\mathcal{R}}_k^\star-\varepsilon_k>\mathcal{R}_k^*,\ k=m\\ \bar{\mathcal{R}}_k^\star+\varepsilon_k>\mathcal{R}_k^*,\ k\neq m\end{cases}. \tag{62}$$

where variables $\varepsilon_k>0$ are sufficiently small. Moverover, the per-antenna power constraint $[\sum_{k=1}^{K}\mathbf{w}_k^\star(\mathbf{w}_k^\star)^H]_{n,n}\leq P_n$ can also hold, where $\mathbf{w}_k^\star=\sqrt{\rho_k^\star}\bar{\mathbf{w}}_k^\star$. Thus

$(\rho_1^\star, \ldots, \rho_m^\star - \varepsilon, \ldots, \rho_K^\star, \bar{\mathbf{w}}_1^\star, \ldots, \bar{\mathbf{w}}_K^\star)$ satisfies all constraints and can achieve lower objective, simultaneously. This is contrary to that $(\rho_1^\star, \ldots, \rho_K^\star, \bar{\mathbf{w}}_1^\star, \ldots, \bar{\mathbf{w}}_K^\star)$ is the optimal solution. Therefore, (61) does not hold and

$$\bar{\mathcal{R}}_m^\star = \mathcal{R}_m^*. \tag{63}$$

The optimal solution of **P2** achieves the same ergodic rates as **P1**, and the optimal solution of **P2** can achieve lower or equal total power while satisfying the per-antenna power constraint. Thus, the equivalence of the optimal solution of **P1** and **P2** is proven.

2) Strong duality and the KKT conditions are necessary and sufficient for the optimal solution (more detailed analysis is given in [3]). In this paper, the optimal structure is derived from the KKT condition $\frac{\partial \mathcal{L}_{\mathcal{R}}}{\partial \mathbf{w}_k} = 0, \forall k \in \mathcal{K}, \forall n \in \mathcal{N}$, where the corresponding Lagrange multipliers are determined by the strong duality between **P3** and its dual problem. Therefore, the optimality of the solution structure can be guaranteed.

Appendix B
Calculation Process for the Expression of $\mathbb{E}\{\mathbf{h}_k \mathbf{H}_k^h\}$

Above all, based on the modified channel model given in (1) and the UPA response vector in (9), the expression of $\mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\}$ can be concretized into

$$\mathbb{E}\left\{\mathbf{h}_k \mathbf{h}_k^H\right\} = \mathbb{E}\left\{|g_k|^2\right\} \mathbb{E}\left\{\mathbf{v}_k \mathbf{v}_k^H\right\} = \sigma_g^2 \mathbb{E}\left\{\mathbf{v}_k \mathbf{v}_k^H\right\}. \tag{64}$$

It is impossible to directly perform the expectation on the matrix $\mathbf{v}_k \mathbf{v}_k^H$, thus we consider the element-wise operation, the n-th element of $\mathbf{v}_k$ can be expressed as

$$\begin{aligned}(\mathbf{v}_k)_n &= \frac{1}{\sqrt{M_x M_y}} e^{-j\pi t \sin(\hat{\theta}_k + \theta_k^e)\cos(\hat{\varphi}_k + \varphi_k^e)} \\ &\quad \times e^{-j\pi(n - tM_y)\cos(\hat{\theta}_k + \theta_k^e)}, \\ n &= 0, \ldots, M_x M_y - 1, t = \lfloor \frac{n}{M_y} \rfloor. \end{aligned} \tag{65}$$

Then the element of m-th row and n-th column of matrix $\mathbb{E}\{\mathbf{h}_k \mathbf{h}_k^H\}$ can be defined as

$$\mathbb{E}\left\{\left[\mathbf{h}_k \mathbf{h}_k^H\right]_{m,n}\right\} = \sigma_g^2 \mathbb{E}\left\{(\mathbf{v}_k)_m (\mathbf{v}_k)_n^H\right\}, \tag{66}$$

where

$$\begin{aligned}&\mathbb{E}\left\{(\mathbf{v}_k)_m (\mathbf{v}_k)_n^H\right\} \\ &= \frac{1}{M_x M_y} \mathbb{E}\left\{e^j \pi (b-a)\sin(\hat{\theta}_k + \theta_k^e)\cos(\hat{\varphi}_k + \varphi_k^e)\right. \\ &\quad \left. + j\pi\left(n - m - (b-a)M_y\right)\cos(\hat{\theta}_k + \theta_k^e)\right\}, \end{aligned} \tag{67}$$

with $m = 0, \ldots, M_x M_y - 1, a = \lfloor \frac{m}{M_y} \rfloor$ and $n = 0, \ldots, M_x M_y - 1, b = \lfloor \frac{n}{M_y} \rfloor$.

A. Uniform Distribution

When φ_k^e and θ_k^e both exhibit a uniform distribution, i.e.,

$$\varphi_k^e \sim U\left[\varphi_L, \varphi_U\right], \tag{68}$$

$$\theta_k^e \sim U\left[\theta_L, \theta_U\right], \tag{69}$$

where φ_L and φ_U are the lower and upper boundary of φ_k^e, while θ_L and θ_U are the lower and upper boundary of θ_k^e, respectively. Then we define Δ_φ for φ_k^e as

$$\Delta_\varphi = \varphi_U - \varphi_L, \tag{70}$$

and Δ_θ for θ_k^e as

$$\Delta_\theta = \theta_U - \theta_L, \tag{71}$$

thus (67) can be further expressed as

$$\begin{aligned}\mathbb{E}\left\{(\mathbf{v}_k)_m (\mathbf{v}_k)_n^H\right\} &= \frac{1}{\Delta_\theta \Delta_\varphi} \frac{1}{M_x M_y} \\ &\quad \times \int_{\theta_L}^{\theta_U} e^{j\pi(n-m-(b-a)M_y)\cos(\hat{\theta}_k + \theta_k^e)} \\ &\quad \times \int_{\varphi_L}^{\varphi_U} e^{j\pi(b-a)\sin(\hat{\theta}_k + \theta_k^e)\cos(\hat{\varphi}_k + \varphi_k^e)} d\varphi_k^e d\theta_k^e. \end{aligned} \tag{72}$$

From (72), we can observe that the expression of $\mathbb{E}\{(\mathbf{v}_k)_m (\mathbf{v}_k)_n^H\}$ can be derived according to the following three cases of $n = m, a = b$; $n \neq m, a = b$ and $n \neq m, a \neq b$. (Because $a = \lfloor \frac{m}{M_y} \rfloor$ and $b = \lfloor \frac{n}{M_y} \rfloor$, $a = b$ must hold when $n = m$.)

- When $n = m, a = b$, we have

$$\mathbb{E}\left\{(\mathbf{v}_k)_m (\mathbf{v}_k)_n^H\right\} = \frac{1}{M_x M_y}; \tag{73}$$

- When $n \neq m, a = b$, (72) can be rewritten as

$$\begin{aligned}\mathbb{E}\left\{(\mathbf{v}_k)_m (\mathbf{v}_k)_n^H\right\} &= \frac{1}{\Delta_\theta} \frac{1}{M_x M_y} \\ &\quad \times \int_{\theta_L}^{\theta_U} e^{j\pi(n-m)\cos(\hat{\theta}_k + \theta_k^e)} d\theta_k^e. \end{aligned} \tag{74}$$

However, (74) is not integrable. It is worth noting that φ_k^e and θ_k^e are random variables that change around zero, with small values. Thus, it is reasonable to let $\cos(\varphi_k^e) \approx 1$ and $\sin(\varphi_k^e) \approx \varphi_k^e$. Without loss of generality, we make the following approximations,

$$\begin{aligned}\cos(\hat{\varphi}_k + \varphi_k^e) &= \cos(\hat{\varphi}_k)\cos(\theta_k^e) - \sin(\hat{\varphi}_k)\sin(\varphi_k^e) \\ &\approx \cos(\hat{\varphi}_k) - \sin(\hat{\varphi}_k)\varphi_k^e, \end{aligned} \tag{75}$$

$$\begin{aligned}\sin(\hat{\theta}_k + \theta_k^e) &= \sin(\hat{\theta}_k)\cos(\theta_k^e) + \cos(\hat{\theta}_k)\sin(\theta_k^e) \\ &\approx \sin(\hat{\theta}_k) + \cos(\hat{\theta}_k)\theta_k^e, \end{aligned} \tag{76}$$

$$\begin{aligned}\cos(\hat{\theta}_k + \theta_k^e) &= \cos(\hat{\theta}_k)\cos(\theta_k^e) - \sin(\hat{\theta}_k)\sin(\theta_k^e) \\ &\approx \cos(\hat{\theta}_k) - \sin(\hat{\theta}_k)\theta_k^e. \end{aligned} \tag{77}$$

Then, (74) can be reformulated as

$$\begin{aligned}\mathbb{E}\left\{(\mathbf{v}_k)_m (\mathbf{v}_k)_n^H\right\} &\approx \frac{1}{\Delta_\theta} \frac{1}{M_x M_y} \\ &\quad \times \int_{\theta_L}^{\theta_U} e^{j\pi(n-m)\left(\cos(\hat{\theta}_k) - \sin(\hat{\theta}_k)\theta_k^e\right)} d\theta_k^e \end{aligned}$$

$$= \frac{j}{\Delta_\theta} \frac{1}{M_x M_y} \left(\frac{e^{j\pi(n-m)\left(\cos(\hat{\theta}_k) - \sin(\hat{\theta}_k)\theta_k^e\right)}}{\pi (n-m) \sin(\hat{\theta}_k)} \right) \Bigg|_{\theta_L}^{\theta_U}. \tag{78}$$

To make the expressions more concise in this paper, we define

$$(F(x))|_l^u = F(u) - F(l), \tag{79}$$

so (78) is equal to

$$\left(\frac{e^{j\pi(n-m)\left(\cos(\hat{\theta}_k) - \sin(\hat{\theta}_k)\theta_k^e\right)}}{\pi (n-m) \sin(\hat{\theta}_k)} \right) \Bigg|_{\theta_L}^{\theta_U} = \frac{e^{j\pi(n-m)\left(\cos(\hat{\theta}_k) - \sin(\hat{\theta}_k)\theta_U\right)}}{\pi (n-m) \sin(\hat{\theta}_k)} - \frac{e^{j\pi(n-m)\left(\cos(\hat{\theta}_k) - \sin(\hat{\theta}_k)\theta_L\right)}}{\pi (n-m) \sin(\hat{\theta}_k)}.$$

For convenience, we let $\theta_L = -\theta_U$ and (78) can be further written as

$$\mathbb{E}\left\{ (\mathbf{v}_k)_m (\mathbf{v}_k)_n^H \right\} = \frac{1}{\Delta_\theta} \frac{1}{M_x M_y} \times \frac{2e^{j\pi(n-m)\cos(\hat{\theta}_k)} \sin\left(\pi (n-m) \sin(\hat{\theta}_k)\theta_U\right)}{\pi (n-m) \sin(\hat{\theta}_k)}. \tag{80}$$

- When $n \neq m, a \neq b$, we first integrate for φ_k^e as,

$$\int_{\varphi_L}^{\varphi_U} e^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\cos(\hat{\varphi}_k+\varphi_k^e)} d\varphi_k^e$$
$$\approx \int_{\varphi_L}^{\varphi_U} e^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)(\cos(\hat{\varphi}_k)-\sin(\hat{\varphi}_k)\varphi_k^e)} d\varphi_k^e$$
$$= \frac{je^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)(\cos(\hat{\varphi}_k))}}{\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\sin(\hat{\varphi}_k)} \times \left(e^{-j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\sin(\hat{\varphi}_k)\varphi_U} - e^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\sin(\hat{\varphi}_k)\varphi_U} \right)$$
$$= \frac{2e^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\cos(\hat{\varphi}_k)}}{\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\sin(\hat{\varphi}_k)} \times \sin\left(\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\sin(\hat{\varphi}_k)\varphi_U\right). \tag{81}$$

where we let $\varphi_L = -\varphi_U$. Based on (81), the integral of θ_k^e can be calculated as

$$\int_{\theta_L}^{\theta_U} 2e^{j\pi(n-m-(b-a)M_y)\cos(\hat{\theta}_k+\theta_k^e)} \times e^{j\pi(b-a)\cos(\hat{\varphi}_k)\sin(\hat{\theta}_k+\theta_k^e)} \frac{\sin\left(B\varphi_U \sin(\hat{\theta}_k+\theta_k^e)\right)}{B\sin(\hat{\theta}_k+\theta_k^e)} d\theta_k^e$$
$$\approx 2Z \int_{\theta_L}^{\theta_U} e^{j\pi A\theta_k^e} \frac{\sin\left(B\varphi_U\left(\sin(\hat{\theta}_k)+\cos(\hat{\theta}_k)\theta_k^e\right)\right)}{B\left(\sin(\hat{\theta}_k)+\cos(\hat{\theta}_k)\theta_k^e\right)} d\theta_k^e, \tag{82}$$

where

$$A = (b-a)\cos(\hat{\varphi}_k)\cos(\hat{\theta}_k) - (n-m-(b-a)M_y)\sin(\hat{\theta}_k), \tag{83}$$
$$B = \pi(b-a)\sin(\hat{\varphi}_k), \tag{84}$$
$$Z = e^{j\pi(n-m-(b-a)M_y)\cos(\hat{\theta}_k)+j\pi(b-a)\cos(\hat{\varphi}_k)\sin(\hat{\theta}_k)}. \tag{85}$$

However, (82) is still not integrable, without loss of generality, we let

$$\frac{\sin\left(B\varphi_U\left(\sin(\hat{\theta}_k)+\cos(\hat{\theta}_k)\theta_k^e\right)\right)}{B\left(\sin(\hat{\theta}_k)+\cos(\hat{\theta}_k)\theta_k^e\right)} \approx \frac{\sin\left(B\varphi_U \sin(\hat{\theta}_k)\right)}{B\sin(\hat{\theta}_k)}, \tag{86}$$

the simulation results have proved the good performance of the above approximation. Then (82) can be calculated as

$$2Z\frac{\sin\left(B\varphi_U\left(\sin(\hat{\theta}_k)\right)\right)}{B\left(\sin(\hat{\theta}_k)\right)} \int_{\theta_L}^{\theta_U} e^{j\pi A\theta_k^e} d\theta_k^e \approx 2Z\frac{\sin\left(B\varphi_U\left(\sin(\hat{\theta}_k)\right)\right)}{B\left(\sin(\hat{\theta}_k)\right)} \frac{2\sin(\pi A\theta_U)}{\pi A} = 4Z\frac{\sin\left(B\varphi_U \sin(\hat{\theta}_k)\right)\sin(\pi A\theta_U)}{\pi AB\sin(\hat{\theta}_k)}, \tag{87}$$

where we also set $\theta_L = -\theta_U$. As a result,

$$\mathbb{E}\left\{ (\mathbf{v}_k)_m (\mathbf{v}_k)_n^H \right\} \approx \frac{1}{M_x M_y}\frac{4Z}{\Delta_\theta\Delta_\varphi} \times \frac{\sin\left(B\varphi_U \sin(\hat{\theta}_k)\right)\sin(\pi A\theta_U)}{\pi AB\sin(\hat{\theta}_k)}. \tag{88}$$

Finally, in the case of uniform distribution, i.e., $\varphi_k^e \sim U(\varphi_L, \varphi_U)$ and $\theta_k^e \sim U(\theta_L, \theta_U)$, the expression of $\mathbb{E}\{\mathbf{h}_k\mathbf{h}_k^H\}$ given in (35) can be obtained.

B. Gaussian Distribution

In this case, we assume θ_k^e is the Gaussian variable with mean $\mu_{\theta,k}$ and variance $\sigma_{\theta,k}^2$, while φ_k^e with mean $\mu_{\varphi,k}$ and variance $\sigma_{\varphi,k}^2$. Thus, the probability density functions of θ_k^e and φ_k^e can be expressed by

$$f\left(\theta_k^e\right) = \frac{1}{\sqrt{2\pi}\sigma_{\theta,k}} e^{\left(-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}\right)}, \tag{89}$$

$$f\left(\varphi_k^e\right) = \frac{1}{\sqrt{2\pi}\sigma_{\varphi,k}} e^{\left(-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}\right)}, \tag{90}$$

so $\mathbb{E}\{(\mathbf{v}_k)_m(\mathbf{v}_k)_n^H\}$ can be rewritten as

$$\begin{aligned}
&\mathbb{E}\left\{(\mathbf{v}_k)_m(\mathbf{v}_k)_n^H\right\}\\
&= \frac{1}{2\pi\sigma_{\varphi,k}\sigma_{\theta,k}}\frac{1}{M_xM_y}\\
&\quad\times\int_{-\infty}^{+\infty} e^{j\pi(n-m-(b-a)M_y)\cos(\hat{\theta}_k+\theta_k^e)} e^{-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}}\\
&\quad\times\int_{-\infty}^{+\infty} e^{j}\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\\
&\quad\times\cos(\hat{\varphi}_k+\varphi_k^e) e^{\left(-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}\right)} d\varphi_k^e d\theta_k^e.
\end{aligned} \tag{91}$$

Similar to the calculation process of the uniform distribution given in Appendix B-A, we discuss the expression of $\mathbb{E}\{\mathbf{h}_k\mathbf{h}_k^H\}$ in the following three cases.

- When $n=m, a=b$, from (91) we obtain

$$\begin{aligned}
&\mathbb{E}\left\{(\mathbf{v}_k)_m(\mathbf{v}_k)_n^H\right\}\\
&= \frac{1}{M_xM_y}\frac{1}{2\pi\sigma_{\varphi,k}\sigma_{\theta,k}}\int_{-\infty}^{+\infty} e^{\left(-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}\right)}\\
&\quad\times\int_{-\infty}^{+\infty} e^{\left(-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}\right)} d\varphi_k^e d\theta_k^e.
\end{aligned} \tag{92}$$

Obviously,

$$\begin{aligned}
&\frac{1}{\sqrt{2\pi}\sigma_{\theta,k}}\int_{-\infty}^{+\infty} e^{-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}} d\theta_k^e\\
&= \frac{1}{2}\,\mathrm{erf}\left(\frac{\theta_k^e-\mu_{\theta,k}}{\sqrt{2}\sigma_{\theta,k}}\right)\Bigg|_{-\infty}^{+\infty} = 1,
\end{aligned} \tag{93}$$

where $\mathrm{erf}(x)$ is the error function defined as

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt. \tag{94}$$

Similarly,

$$\frac{1}{\sqrt{2\pi}\sigma_{\varphi,k}}\int_{-\infty}^{+\infty} e^{\left(-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}\right)} d\varphi_k^e = 1. \tag{95}$$

Thus,

$$E\left\{(\mathbf{v}_k)_m(\mathbf{v}_k)_n^H\right\} = \frac{1}{M_xM_y}. \tag{96}$$

- When $n\neq m, a=b$, we have

$$\begin{aligned}
&\mathbb{E}\left\{(\mathbf{v}_k)_m(\mathbf{v}_k)_n^H\right\}\\
&= \frac{1}{M_xM_y}\int_{-\infty}^{+\infty}\frac{1}{\sqrt{2\pi}\sigma_{\theta,k}}\\
&\quad\times e^{j\pi(n-m)\cos(\hat{\theta}_k+\theta_k^e)} e^{-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}} d\theta_k^e\\
&\approx \frac{1}{M_xM_y}\int_{-\infty}^{+\infty}\frac{1}{\sqrt{2\pi}\sigma_{\theta,k}}\\
&\quad\times e^{j\pi(n-m)\left(\cos(\hat{\theta}_k)-\sin(\hat{\theta}_k)\theta_k^e\right)} e^{-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}} d\theta_k^e\\
&= \frac{e^{j\pi(n-m)\cos(\hat{\theta}_k)}}{M_xM_y}\int_{-\infty}^{+\infty}\frac{1}{\sqrt{2\pi}\sigma_{\theta,k}}\\
&\quad\times e^{-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}-j\pi(n-m)\sin(\hat{\theta}_k)\theta_k^e} d\theta_k^e,
\end{aligned} \tag{97}$$

with

$$\begin{aligned}
&\int_{-\infty}^{+\infty}\frac{1}{\sqrt{2\pi}\sigma_{\theta,k}} e^{-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}-j\pi(n-m)\sin(\hat{\theta}_k)\theta_k^e} d\theta_k^e\\
&= -\frac{1}{2} e^{-j\pi(n-m)\mu_{\theta,k}\sin(\hat{\theta}_k)-\frac{1}{2}\pi^2(n-m)^2\sigma_{\theta,k}^2\sin^2(\hat{\theta}_k)}\\
&\quad\times\mathrm{erf}\left(\frac{-j\pi(n-m)\sin(\hat{\theta}_k)\sigma_{\theta,k}^2+\mu_{\theta,k}-\theta_k^e}{\sqrt{2}\sigma_{\theta,k}}\right)\Bigg|_{-\infty}^{+\infty}\\
&\approx e^{-j\pi(n-m)\mu_{\theta,k}\sin(\hat{\theta}_k)-\frac{1}{2}\pi^2(n-m)^2\sigma_{\theta,k}^2\sin^2(\hat{\theta}_k)}.
\end{aligned} \tag{98}$$

Substituting (98) into (97), we obtain

$$\begin{aligned}
&\mathbb{E}\left\{(\mathbf{v}_k)_m(\mathbf{v}_k)_n^H\right\}\\
&\approx \tfrac{1}{M_xM_y} e^{j\pi(n-m)\cos(\hat{\theta}_k)}\\
&\qquad\times e^{-j\pi(n-m)\mu_{\theta,k}\sin(\hat{\theta}_k)-\frac{1}{2}\pi^2(n-m)^2\sigma_{\theta,k}^2\sin^2(\hat{\theta}_k)}\\
&= \tfrac{1}{M_xM_y} e^{jQ-jP\mu_{\theta,k}-\frac{1}{2}P^2\sigma_{\theta,k}^2},
\end{aligned} \tag{99}$$

where $P{=}\pi(n-m)\sin(\hat{\theta}_k)$ and $Q{=}\pi(n-m)\cos(\hat{\theta}_k)$.

- When $n\neq m, a\neq b$, the integral of φ_k^e is calculated first in the following. From (91), we have

$$\begin{aligned}
&\int_{-\infty}^{+\infty} e^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\cos(\hat{\varphi}_k+\varphi_k^e)}\\
&\quad\times\frac{1}{\sqrt{2\pi}\sigma_{\varphi,k}} e^{-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}} d\varphi_k^e\\
&\approx\int_{-\infty}^{+\infty} e^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\left(\cos(\hat{\varphi}_k)-\sin(\hat{\varphi}_k)\varphi_k^e\right)}\\
&\quad\times\frac{1}{\sqrt{2\pi}\sigma_{\varphi,k}} e^{-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}} d\varphi_k^e\\
&= e^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\cos(\hat{\varphi}_k)}\int_{-\infty}^{+\infty}\\
&\quad\times\frac{1}{\sqrt{2\pi}\sigma_{\varphi,k}} e^{-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}+j\pi A\varphi_k^e} d\varphi_k^e,
\end{aligned} \tag{100}$$

where $A=-(b-a)\sin(\hat{\theta}_k+\theta_k^e)\sin(\hat{\varphi}_k)$ and the expression of $\int_{-\infty}^{+\infty}\frac{1}{\sqrt{2\pi}\sigma_{\varphi,k}}e^{-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}+j\pi A\varphi_k^e}d\varphi_k^e$ can be further calculated by

$$\begin{aligned}&\int_{-\infty}^{+\infty}\frac{1}{\sqrt{2\pi}\sigma_{\varphi,k}}e^{\left(-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}+j\pi A\varphi_k^e\right)}d\varphi_k^e\\&=-\frac{1}{2}e^{j\pi A\mu_{\varphi,k}-\frac{1}{2}\pi^2A^2\sigma_{\varphi,k}^2}\operatorname{erf}\\&\times\left(\sqrt{2}\frac{\left(j\pi A\sigma_{\varphi,k}^2+\mu_{\varphi,k}-\varphi_k^e\right)}{2\sigma_{\varphi,k}}\right)\Bigg|_{-\infty}^{+\infty}\\&=e^{j\pi A\mu_{\varphi,k}-\frac{1}{2}\pi^2A^2\sigma_{\varphi,k}^2}.\end{aligned}\tag{101}$$

Substitude the expression of A and (101) into (100), we then obtain

$$\begin{aligned}&\int_{-\infty}^{+\infty}e^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\cos(\hat{\varphi}_k+\varphi_k^e)}\\&\quad\times\frac{1}{\sqrt{2\pi}\sigma_{\varphi,k}}e^{-\frac{(\varphi_k^e-\mu_{\varphi,k})^2}{2\sigma_{\varphi,k}^2}}d\varphi_k^e\\&\quad=e^{j\pi(b-a)\sin(\hat{\theta}_k+\theta_k^e)\cos(\hat{\varphi}_k)}e^{j\pi A\mu_{\varphi,k}-\frac{1}{2}\pi^2A^2\sigma_{\varphi,k}^2}\\&\quad=e^{j(\pi(b-a)\cos(\hat{\varphi}_k)-\pi(b-a)\sin(\hat{\varphi}_k)\mu_{\varphi,k})\sin(\hat{\theta}_k+\theta_k^e)}\\&\quad\quad\times e^{-\frac{1}{2}\pi^2\sigma_{\varphi,k}^2((b-a)\sin(\hat{\varphi}_k))^2\left(\sin(\hat{\theta}_k+\theta_k^e)\right)^2}\\&\quad=e^{jD\sin(\hat{\theta}_k+\theta_k^e)-\frac{1}{2}F\left(\sin(\hat{\theta}_k+\theta_k^e)\right)^2},\end{aligned}\tag{102}$$

where $D=\pi(b-a)\cos(\hat{\varphi}_k)-\pi(b-a)\sin(\hat{\varphi}_k)\mu_{\varphi,k}$ and $F=\pi^2\sigma_{\varphi,k}^2((b-a)\sin(\hat{\varphi}_k))^2$.

Based on (102), (91) can be simplified into

$$\begin{aligned}&\mathbb{E}\left\{(\mathbf{v}_k)_m(\mathbf{v}_k)_n^H\right\}\\&\quad=\frac{1}{M_xM_y}\frac{1}{\sqrt{2\pi}\sigma_{\theta,k}}\int_{-\infty}^{+\infty}e^{j\pi(n-m-(b-a)M_y)\cos(\hat{\theta}_k+\theta_k^e)}\\&\quad\times e^{\left(-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}\right)+jD\sin(\hat{\theta}_k+\theta_k^e)-\frac{1}{2}F\left(\sin(\hat{\theta}_k+\theta_k^e)\right)^2}d\theta_k^e.\end{aligned}\tag{103}$$

To enable (103) to be integrated, we perform the following approximation,

$$\left(\sin(\hat{\theta}_k+\theta_k^e)\right)^2\approx\frac{1-\cos(2\hat{\theta}_k)+2\sin(2\hat{\theta}_k)\theta_k^e}{2}.\tag{104}$$

Here, we let $C=jD\cos(\hat{\theta}_k)-\frac{1}{2}F\sin(2\hat{\theta}_k)-j\pi(n-m-(b-a)M_y)\sin(\hat{\theta}_k)$ and $E=jD\sin(\hat{\theta}_k)-\frac{1}{4}F(1-\cos(2\hat{\theta}_k))+j\pi(n-m-(b-a)M_y)\cos(\hat{\theta}_k)$, and then (103) can be further expressed as

$$\begin{aligned}&\mathbb{E}\left\{(\mathbf{v}_k)_m(\mathbf{v}_k)_n^H\right\}=\frac{1}{M_xM_y}e^E\int_{-\infty}^{+\infty}\frac{1}{\sqrt{2\pi}\sigma_{\theta,k}}\\&\qquad\times e^{-\frac{(\theta_k^e-\mu_{\theta,k})^2}{2\sigma_{\theta,k}^2}+C\theta_k^e}d\theta_k^e\\&\quad=-\frac{1}{2}\frac{1}{M_xM_y}e^{\frac{1}{2}C^2\sigma_{\theta,k}^2+C\mu_{\theta,k}+E}\operatorname{erf}\\&\quad\times\left(\frac{\sqrt{2}\left(C\sigma_{\theta,k}^2+\mu_{\theta,k}\right)-\theta_k^e}{2\sigma_{\theta,k}}\right)\Bigg|_{-\infty}^{+\infty}\\&\quad=\frac{1}{M_xM_y}e^{\left(\frac{1}{2}C^2\sigma_{\theta,k}^2+C\mu_{\theta,k}\right)+E}.\end{aligned}\tag{105}$$

As a result, combined with (96), (99) and (105), we obtain (39) holds finally.

Appendix C
The Detailed Description and Modeling of the DL Network

As illustrated in Fig. 3, we model the DL network in the structural order.

- 1) *Input Layer:* The input layer processes the input information and transmits the results to the subsequent convolutional layer groups for feature extraction. In this paper, the input information is $\bar{\mathbf{H}}\in\mathbb{C}^{N_T\times N_T\times 2K}$, which is denoted by (46), (47) and (49).
- 2) *Convolutional Layer Groups:* Assume that the DL network contains $\mathcal{C}$ CL groups, consider the $c\in\mathcal{C}$-th CL group.
- 2.1) *CL:* The CL has n_c kernels, each kernel has a size of $k_c^h\times k_c^w\times k_c^l$. The input of the c-th CL group is given by $\mathbf{\Lambda}_{conv,c}\in\mathbb{R}^{\alpha_{c-1}\times\beta_{c-1}\times\kappa_{c-1}\times n_{c-1}}$, where $\alpha_{c-1},\beta_{c-1},\kappa_{c-1}$ denote the size of each feature obtained by the c-1-th CL group. The output of the CL can be expressed as

$$\mathbf{\Pi}_{conv,c}=Conv\left(\mathbf{\Lambda}_{conv,c},\mathbf{\Omega}_c,\mathbf{\Theta}_c\right),\quad c\in\mathcal{C}\tag{106}$$

  where $\mathbf{\Omega}_c$ and $\mathbf{\Theta}_c$ denote the weights and bias vectors of the CL, respectively. $\mathbf{\Pi}_{conv,c}\in\mathbb{R}^{\alpha_c\times\beta_c\times\kappa_c\times n_c}$ will be the input of the BN.
- 2.2) *BN:* The purpose of introducing the BN is to accelerate the convergence speed and enhance the generalization ability of the DL network. The BN normalizes the output of the CL in the following way

$$\begin{aligned}&\mathbf{B}_{bn,c,s}[i,j,k]=\frac{\mathbf{\Pi}_{conv,c,s}[i,j,k]-\mu_{c,s}}{\sqrt{\sigma_{c,s}+\varepsilon}},c\in\mathcal{C}\\&s=1,\ldots,n_c,\ i=1,\ldots,\alpha_c,\ j=1,\ldots,\beta_c,\ k=1,\ldots,\kappa_c\end{aligned}\tag{107}$$

  where $\mathbf{\Pi}_{conv,c,s}\in\mathbb{R}^{\alpha_c\times\beta_c\times\kappa_c}$ denotes the s-th slice of the matrix $\mathbf{\Pi}_{conv,c}$. ε is an independent small quantity. $\mu_{c,s}$ and $\sigma_{c,s}$ denote the batch mean and batch standard deviation, respectively. When the size of each batch is N, we have

$$\mu_{c,s}=\frac{\sum_{n=1}^{N}\sum_{i=1}^{\alpha_c}\sum_{j=1}^{\beta_c}\sum_{k=1}^{\kappa_c}\mathbf{\Pi}_{conv,c,s,n}[i,j,k]}{N\alpha_c\beta_c\kappa_c}\tag{108}$$

$$\sigma_{c,s}=\frac{\begin{array}{c}\sum_{n=1}^{N}\sum_{i=1}^{\alpha_c}\sum_{j=1}^{\beta_c}\sum_{k=1}^{\kappa_c}\\\times\left(\mathbf{\Pi}_{conv,c,s,n}[i,j,k]-\mu_{c,s}\right)^2\end{array}}{N\alpha_c\beta_c\kappa_c}\tag{109}$$

where $\mathbf{\Pi}_{conv,c,s,n}$ denotes the n-th sample in the batch. The BN does not change the size of the input, i.e., $\mathbf{B}_{bn,c} \in \mathbb{R}^{\alpha_c \times \beta_c \times \kappa_c \times n_c}$.

- 2.3) *AC:* The sigmoid function is applied in the AC, which is given by (48). The output of the AC is given by

$$\mathbf{A}_{ac,c} = \text{sigmoid}(\mathbf{B}_{bn,c}) \tag{110}$$

The AC does not change the size of the input, i.e., $\mathbf{A}_{ac,c} \in \mathbb{R}^{\alpha_c \times \beta_c \times \kappa_c \times n_c}$.

- 3) *Fully-Connected Layer Group:* The DL network contains one FC layer group.
- 3.1) *Flatten Layer*. The flatten layer is used to transform the input into a vector $\mathbf{F}_l \in \mathbb{R}^{\alpha_C \beta_C \kappa_C n_C}$, i.e.,

$$\mathbf{F}_f = flat(\mathbf{A}_{ac,C}) \tag{111}$$

where $\mathbf{A}_{ac,C}$ is the output of the last CL group.

- 3.2) *FC:* The FC layer maps the information obtained by CL groups to the key features. Assume that there are $\mathcal{H}$ hidden FC layers. Consider the $h \in \mathcal{H}$-th hidden layer, which has m_h neurals. The output of the h-th hidden layer $\mathbf{Fc}_h \in \mathbb{R}^{m_h}$ is given by

$$\mathbf{Fc}_h = \mathbf{Fc}_{h-1}\mathbf{W}_h + \mathbf{b}_h \tag{112}$$

where $\mathbf{W}_h \in \mathbb{R}^{m_{h-1} \times m_h}$ and $\mathbf{b} \in \mathbb{R}^{m_h}$ denote the weight matrix and bias vector of the h-th hidden layer, respectively. Besides, $m_1 = \alpha_C \beta_C \kappa_C n_C$.

- 3.3) *Output Layer:* The output layer maps the key features obtained by hidden FC layers to the prediction target $\mathbf{T} \in \mathbb{R}^t$, which is given by

$$\mathbf{T} = \mathbf{Fc}_H \mathbf{W}_t + \mathbf{b}_t \tag{113}$$

where t denotes the number of neurals, $\mathbf{W}_t \in \mathbb{R}^{m_H \times m_t}$ and $\mathbf{b} \in \mathbb{R}^{m_t}$ denote the weight matrix and bias vector of the output layer, respectively.

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Yanhao Liu received the B.E. degree in information engineering from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2019 and the M.E. degree with National Mobile Communications Research Laboratory, Southeast University, Nanjing, China, in 2022. His research interests include wireless communication and signal processing, with the emphasis on satellite communication systems.

Yibiao Wang received the B.E. and M.E. degrees in electronic engineering from Southeast University, Nanjing, China, in 2019 and 2022, respectively. His research interests include precoding, beamforming, and channel prediction, with a focus on massive MIMO systems and machine learning.

Jue Wang (Member, IEEE) received the B.S. degree in communications engineering from Nanjing University, Nanjing, China, in 2006, and the M.S. and Ph.D. degrees from National Mobile Communications Research Laboratory, Southeast University, Nanjing, China, in 2009 and 2014, respectively. From 2014 to 2016, he was a Postdoctoral Research Fellow with the Singapore University of Technology and Design, Singapore. He is currently with the School of Information Science and Technology, Nantong University, Nantong, China. His research interests include MIMO/Massive MIMO wireless communications and machine learning in communication systems. He was the Technical Program Committee Member for a number of IEEE conferences and a Reviewer for various IEEE journals. He was the recipient of the Award Exemplary Reviewers of IEEE TRANSACTIONS ON COMMUNICATIONS in 2014 and IEEE WIRELESS COMMUNICATIONS LETTERS in 2021.

Li You (Senior Member, IEEE) received the B.E. and M.E. degrees in electrical engineering from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2009 and 2012, respectively, and the Ph.D. degree in electrical engineering from Southeast University, Nanjing, in 2016.

From 2014 to 2015, he was a Visiting Researcher with the Center for Pervasive Communications and Computing, University of California Irvine, Irvine, CA, USA. Since 2016, he has been with the Faculty of the National Mobile Communications Research Laboratory, Southeast University, Nanjing, China. His research interests include the general areas of communications, signal processing, and information theory, with the current emphasis on massive MIMO and XL-MIMO communications.

He was the recipient of the National Excellent Doctoral Dissertation Award from the China Institute of Communications, in 2017, the Young Elite Scientists Sponsorship Program (2019–2021) by the China Association for Science and Technology, and the URSI Young Scientist Award in 2021.

Wenjin Wang (Member, IEEE) received the Ph.D. degree in communication and information systems from Southeast University, Nanjing, China, in 2011. From 2010 to 2014, he was with the School of System Engineering, University of Reading, Reading, U.K. He is currently a Professor with the National Mobile Communications Research Laboratory, Southeast University, Nanjing, China. His current research interests include advanced signal processing for future wireless communications and satellite communications. He was the recipient of the Best Paper Award at IEEE WIRELESS COMMUNICATIONS AND SIGNAL PROCESSING, 2009 and the first grade Technological Invention Award of the State Education Ministry of China in 2009.

Xiqi Gao (Fellow, IEEE) received the Ph.D. degree in electrical engineering from Southeast University, Nanjing, China, in 1997. In April 1992, he joined the Department of Radio Engineering, Southeast University. From September 1999 to August 2000, he was a Visiting Scholar with the Massachusetts Institute of Technology, Cambridge, MA, USA and Boston University, Boston, MA, USA. Since May 2001, he has been a Professor of information systems and communications. From August 2007 to July 2008, he was a Humboldt Scholar with the Darmstadt University of Technology, Darmstadt, Germany. His current research interests include broadband multicarrier communications, massive MIMO wireless communications, satellite communications, optical wireless communications, information theory, and signal processing for wireless communications. From 2007 to 2012, he was the Editor for the IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS. From 2009 to 2013, he was an Associate Editor for the IEEE TRANSACTIONS ON SIGNAL PROCESSING. From 2015 to 2017, he was the Editor for the IEEE TRANSACTIONS ON COMMUNICATIONS. He was the recipient of the Science and Technology Awards of the State Education Ministry of China in 1998, 2006, and 2009, the National Technological Invention Award of China, in 2011, the Science and Technology Award of Jiangsu Province of China, in 2014, and the 2011 IEEE Communications Society, Stephen O. Rice Prize Paper Award in the field of communications theory.